

Lecture 1: Introduction to BIP

Dan Wu

College of Biomedical Engineering and Instrumental Science

danwu.bme@zju.edu.cn

Contents

- **Digital Images and Biomedical Images**
- **Format of Digital Images**
- **What is Image Processing**
- **Human Vision System**

What is an Image?

Representation of something else

Objects

Images

Pictures



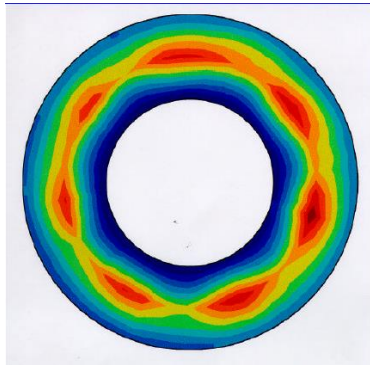
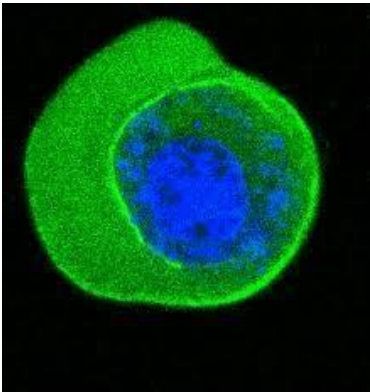
Photographs



Paintings

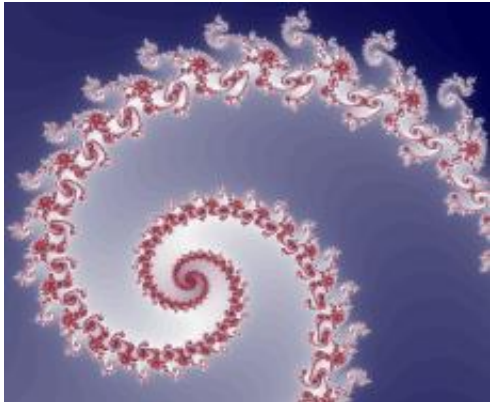
Visible Images

Optical images



Non-visible
physical images

Mathematical functions



What is a Digital Image?

❖ A numerical representation of an object

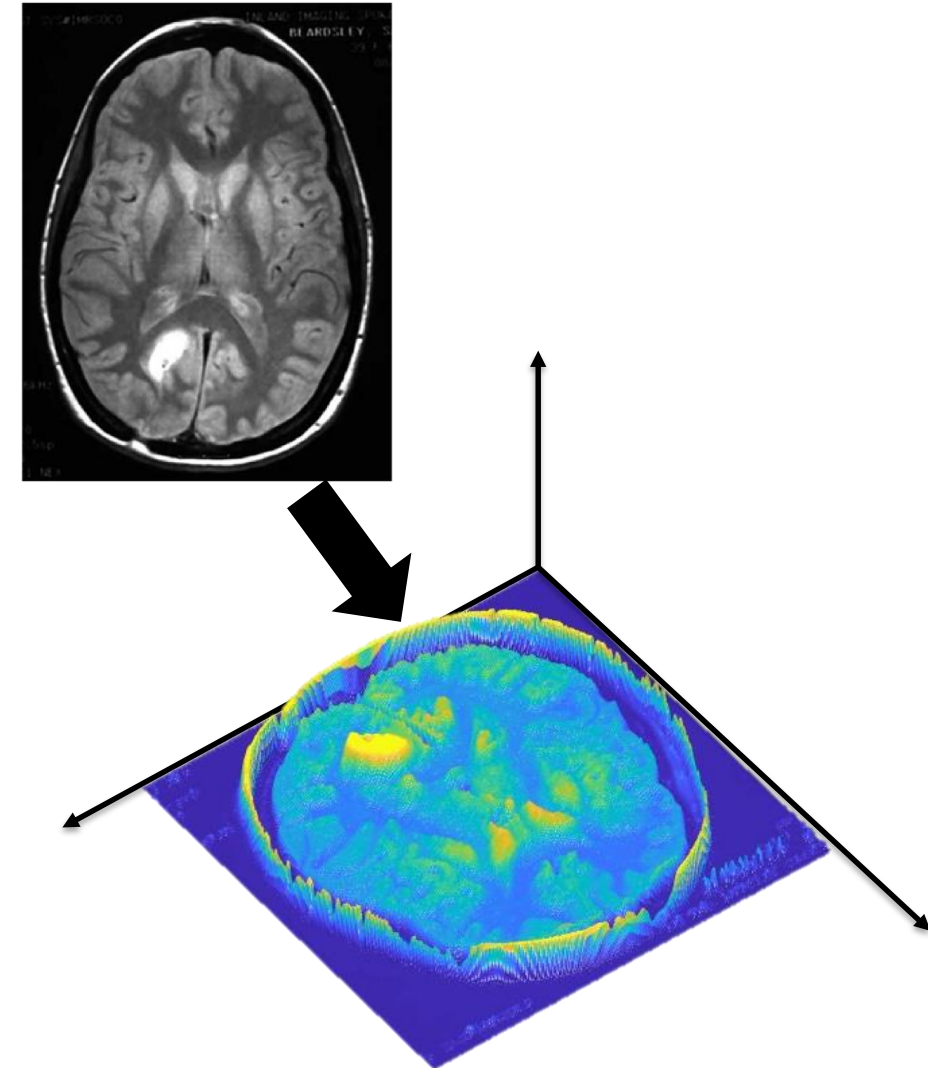
❖ In other way:

An image $\rightarrow f(x, y)$, a 2-D function

$x, y \rightarrow$ spatial (plane) coordinates

$f \rightarrow$ intensity (gray level) at (x, y)

A digital image $\rightarrow x, y, f$ are all finite,
discrete quantities



What is a Digital Image?

- ❖ A numerical representation of an object
- ❖ In a digital image, $f(x, y)$ are called *elements*, also referred as: *picture elements*, or *pixels*
If a 3D digital image (typical medical image): *voxels*

2	2	2	2	0	2	0
2	2	2	0	0	0	0
2	2	4	2	4	4	4
2	4	8	8	10	12	12
2	6	10	12	12	14	18
2	4	8	4	4	2	10
2	2	4	8	8	0	2
2	2	4	8	12	8	2

Pixel: (pixel=picture element)



History of Digital Image

- The first application of digital images was in the newspaper industry, when pictures were first **sent by submarine cable** between London and New York across the Atlantic in early 1921.
- From more than a week shipping to in **less than three hours**
- The initial problems in improving the visual quality of this early digital pictures were related to the
 - selection of printing procedures
 - the distribution of intensity levels



History of Digital Image

1921



1929

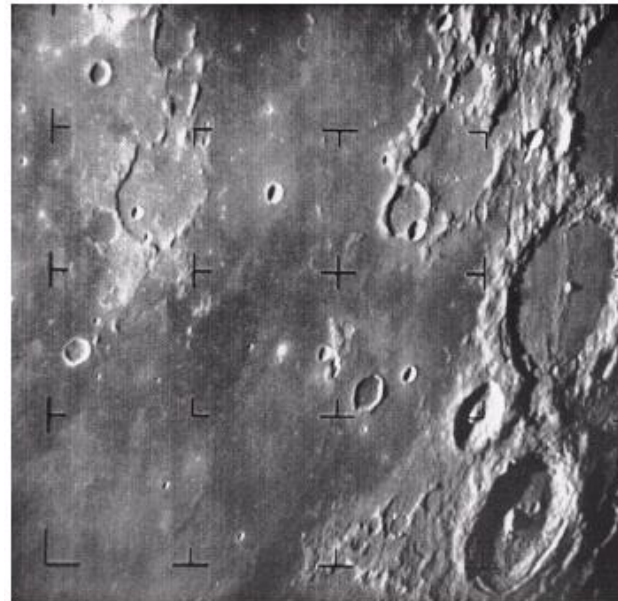


Coded image in 5 distinct levels of gray → 15 levels

The real history is intimately tied to the development of the computer from 1940s

History of Digital Image

The first powerful computer in 1960s and the onset of the space program meanwhile made the birth of what we call digital image processing today. *A wonderful combination!*



Images were processed to correct various types of distortion inherent in the on-board television camera, then came the enhancement and restoration.

History of Digital Image

- Remote earth resources observation
- Pollution pattern analysis, weather prediction
- Archeology
- Physics – high-energy plasmas and electron microscopy
- Astronomy
- Biology
- Nuclear medicine
- Law enforcement
- Defense – military recognizance
- Industrial applications – product assembly and inspection

Today, there is almost **no area of technical endeavor that is not impacted in some way by digital image processing.**

Examples of Fields that Use Digital Images

- The principal energy source for images in use today is the electromagnetic (EM) energy spectrum.

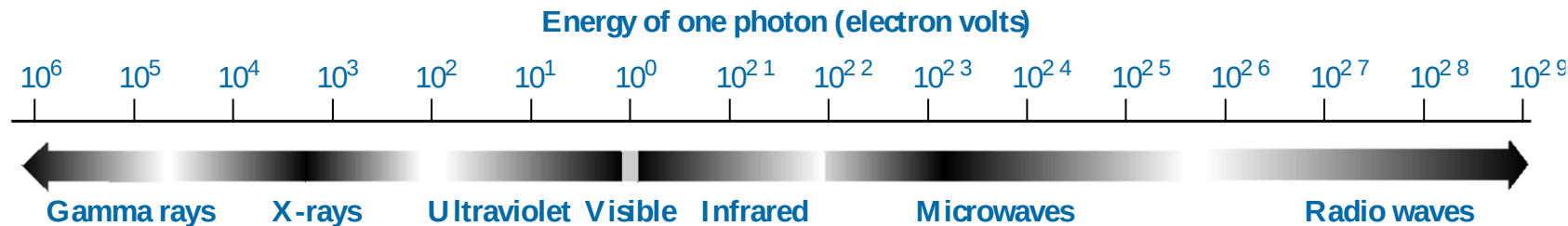
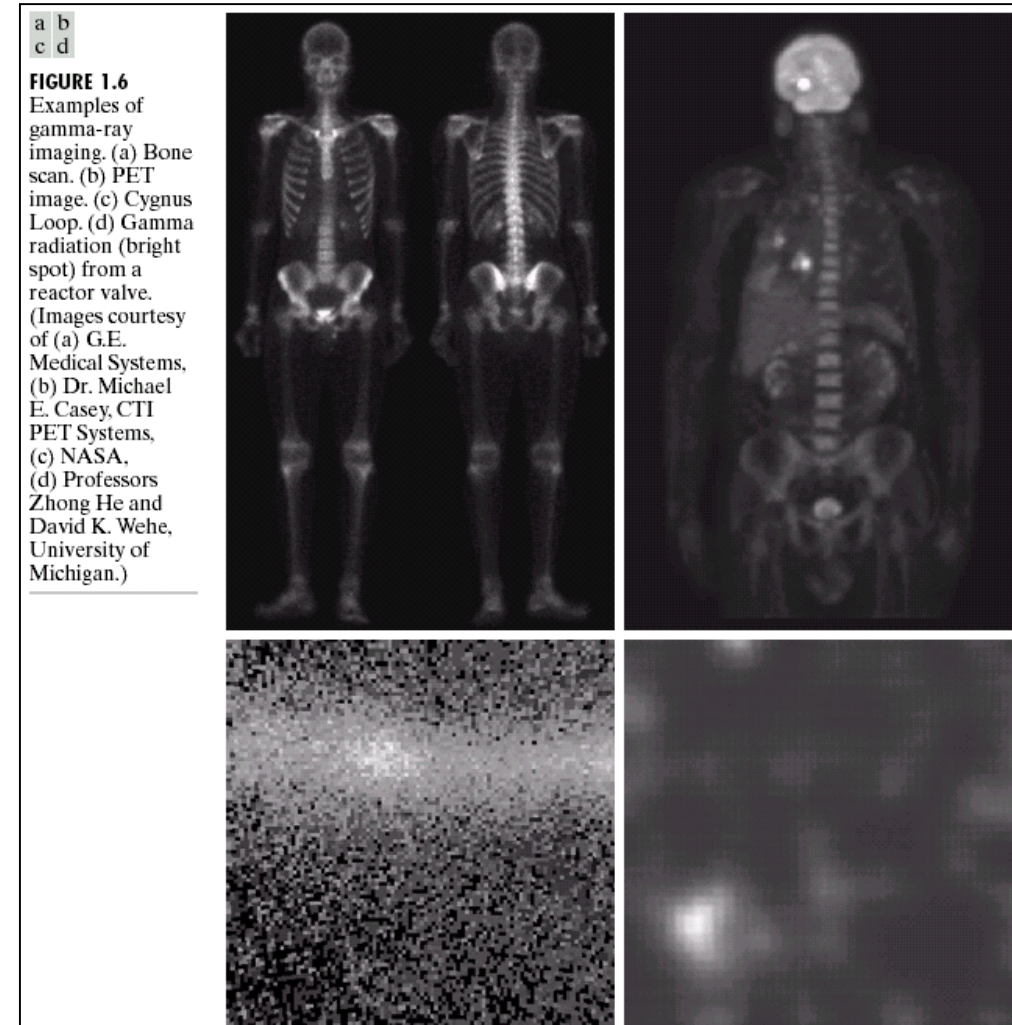


FIGURE 1.5 The electromagnetic spectrum arranged according to energy per photon.

- Other important sources of energy include acoustic, ultrasonic, and electronic (electron beams).
- Synthetic images, used for modeling and visualization, are generated by computer.

EM spectrum: Gamma-Ray Imaging

- Two major uses of gamma rays imaging: **nuclear medicine (PET)** and astronomical observations.
- The first uses radioactive isotope that emits gamma ray as it decays, the second, however, uses the natural radiation of the object being imaged.

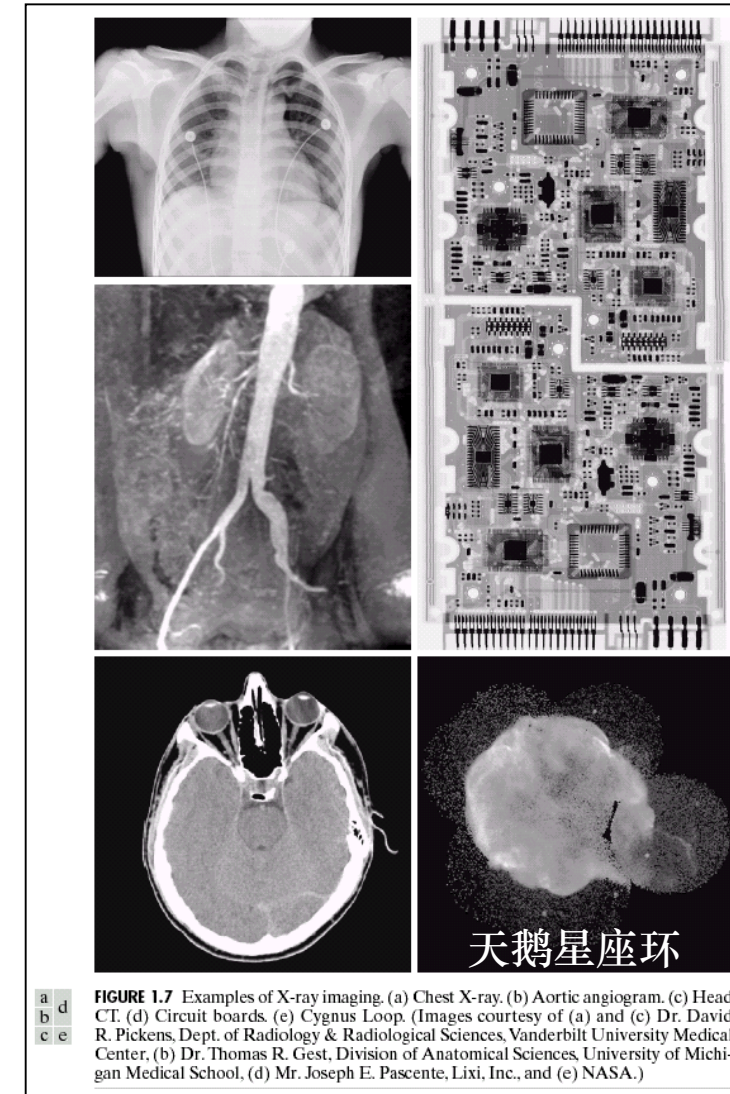


天鹅星座环

核反应堆阀门

EM spectrum: X-Ray Imaging

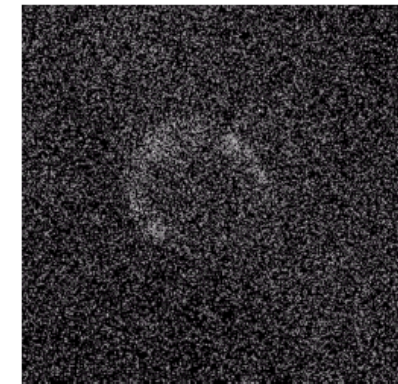
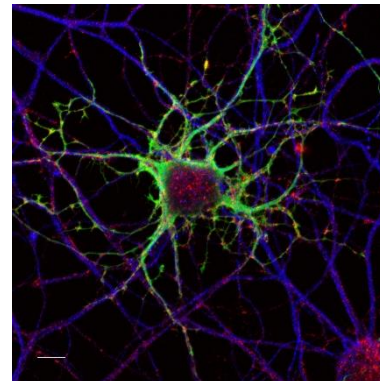
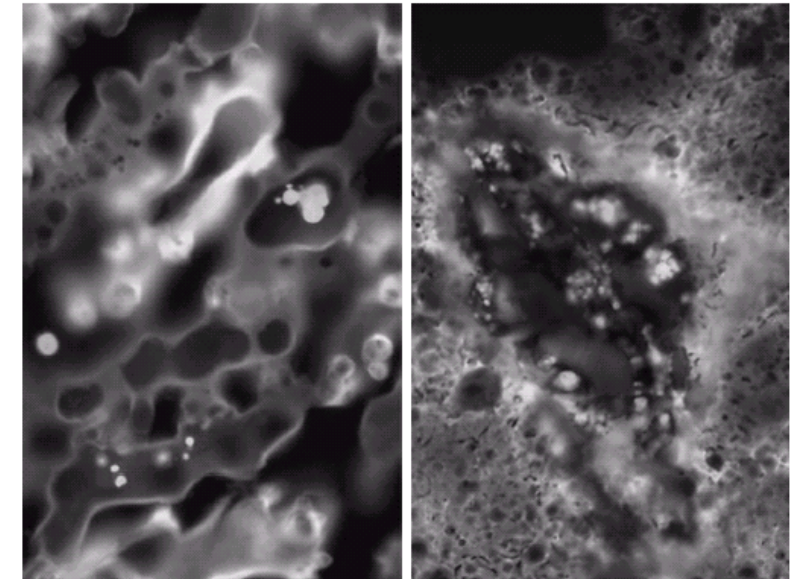
- X-rays are among the oldest sources of EM radiation used for imaging. The best known use is medical diagnostics.
 - Medical CT
 - Digital radiography
 - Angiography
 - Industrial CT
- Also used extensively in industry and other areas like astronomy.



EM spectrum: Ultraviolet Band

- Applications of ultraviolet light include industrial inspection, microscopy, lasers, biological imaging, and astronomical observations.
- The fastest growing areas of microscopy is **fluorescence** microscopy, in which the ultraviolet light is used.

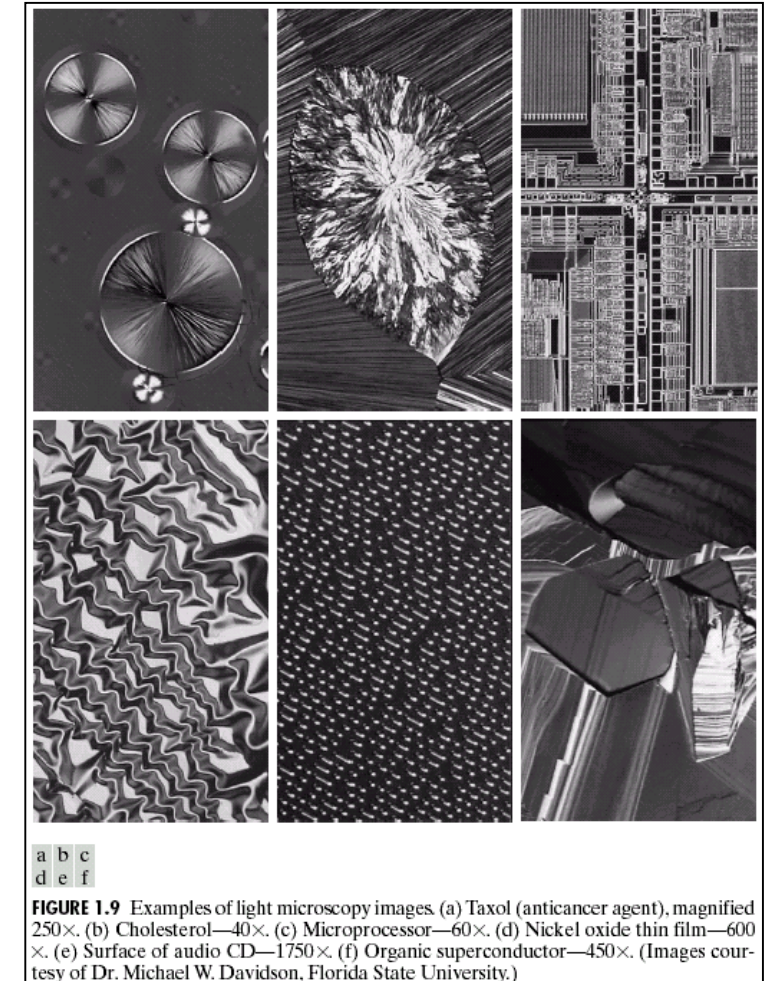
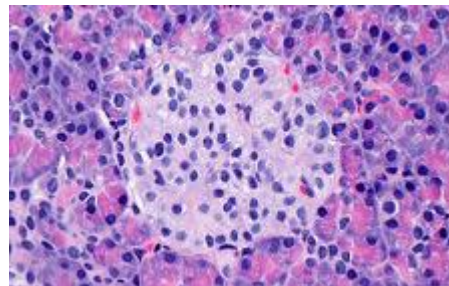
a b
c
FIGURE 1.8
Examples of
ultraviolet
imaging.
(a) Normal corn.
(b) Smut corn.
(c) Cygnus Loop.
(Images courtesy
of (a) and
(b) Dr. Michael
W. Davidson,
Florida State
University,
(c) NASA.)



天鹅星座环

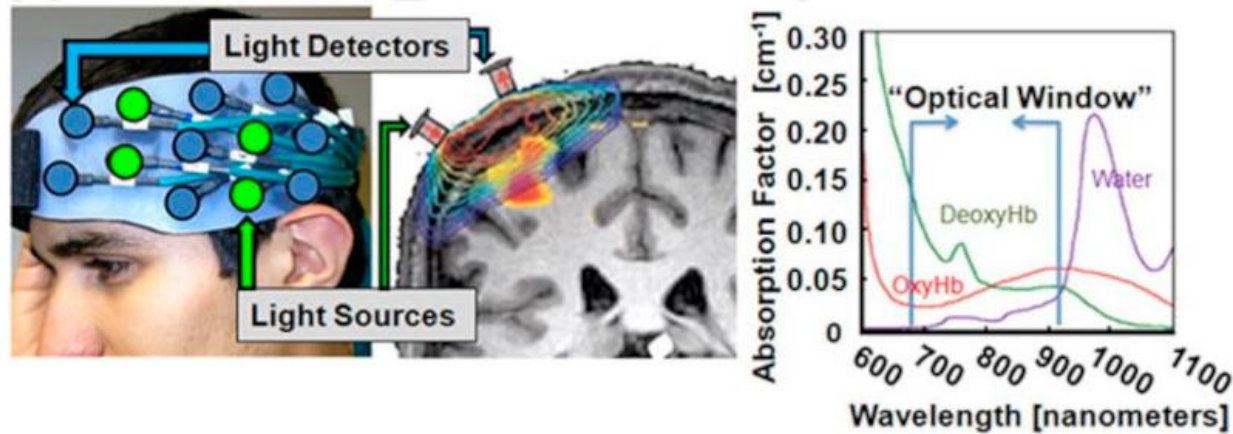
EM spectrum: Visible and Infrared Band

- Imaging in the **visible band** outweighs by far all the others in terms of scope of application.
- The **infrared band** often is used in conjunction with visual imaging.
- Major applications are in:
 - **light microscopy**
 - astronomy
 - remote sensing
 - industry
 - law enforcement



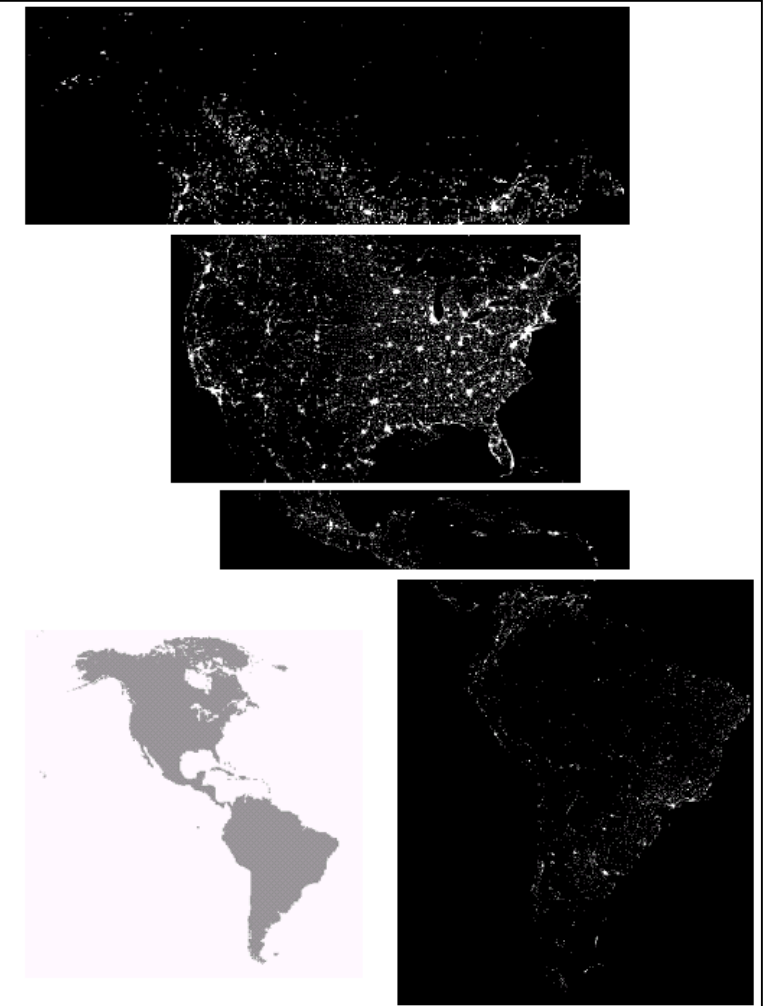
EM spectrum: Visible and Infrared Band

- Part of the Nighttime of the world data set, provides a global inventory of human settlements and Hbo level
- The infrared imaging system operates in the band $10.0\text{-}13.4\mu\text{m}$



Measures changes in oxygenated and deoxygenated hemoglobin concentrations

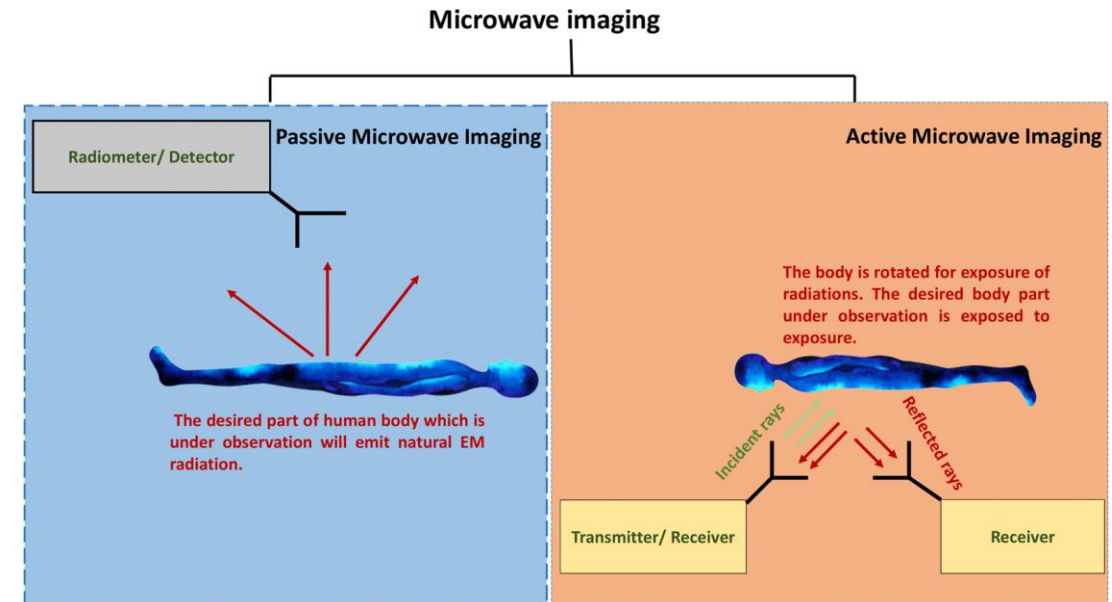
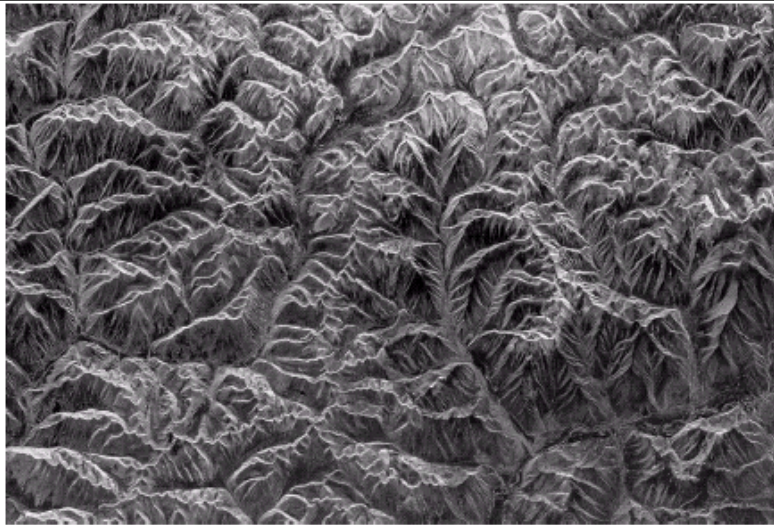
FIGURE 1.12
Infrared satellite
images of the
Americas. The
small gray map is
provided for
reference.
(Courtesy of
NOAA.)



EM spectrum: Microwave Band

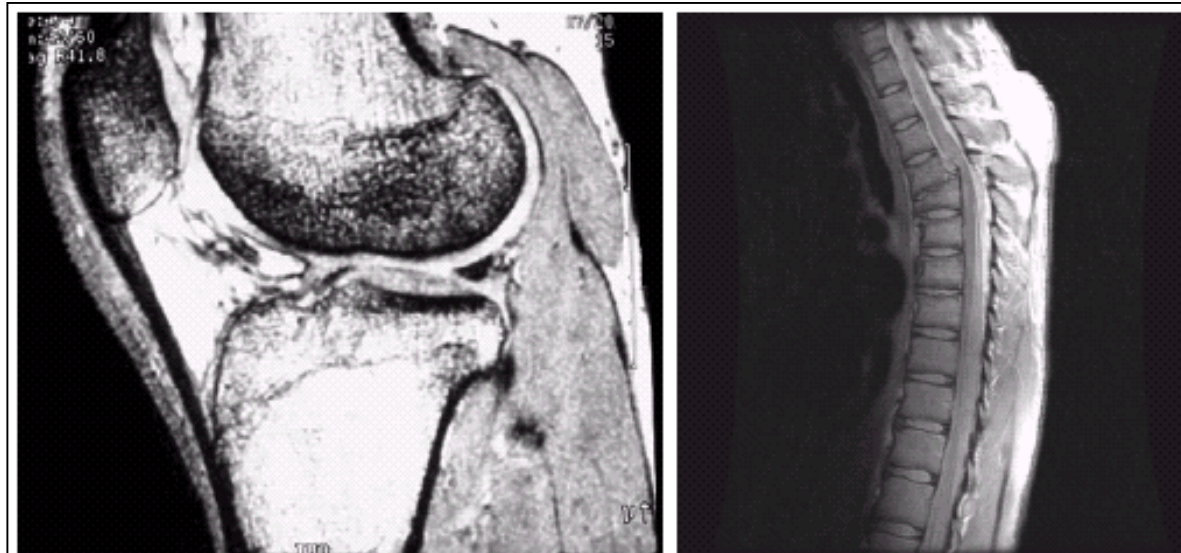
- The dominant application of imaging in the microwave band is **radar**.

FIGURE 1.16
Spaceborne radar
image of
mountains in
southeast Tibet.
(Courtesy of
NASA.)



EM spectrum: Radio Band

- The major applications of imaging in the radio band are in MRI and astronomy



a b

FIGURE 1.17 MRI images of a human (a) knee, and (b) spine. (Image (a) courtesy of Dr. Thomas R. Gest, Division of Anatomical Sciences, University of Michigan Medical School, and (b) Dr. David R. Pickens, Department of Radiology and Radiological Sciences, Vanderbilt University Medical Center.)

From Gamma to Radio: Totally Different View

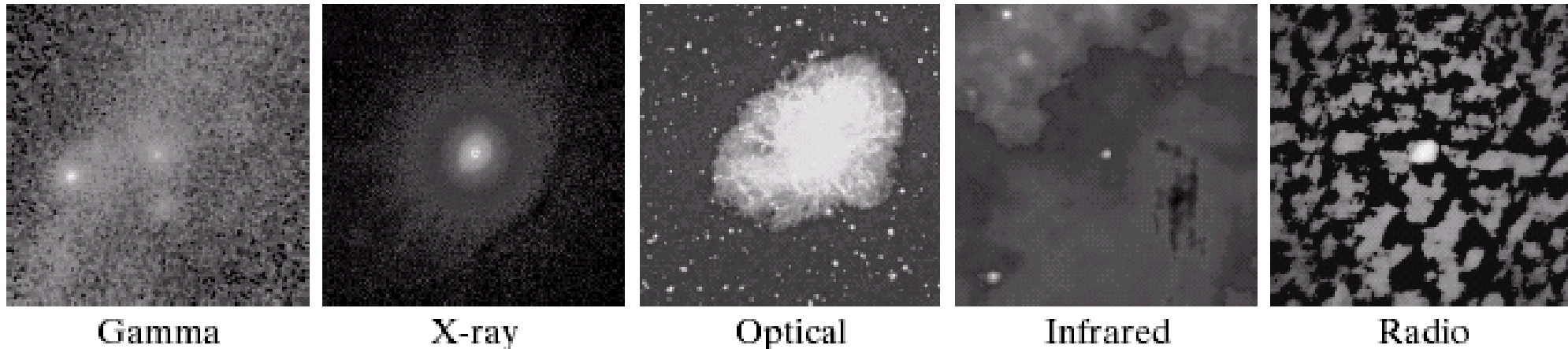
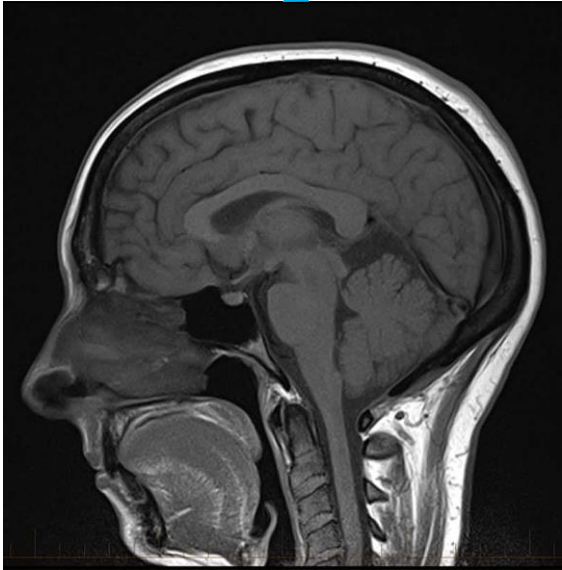
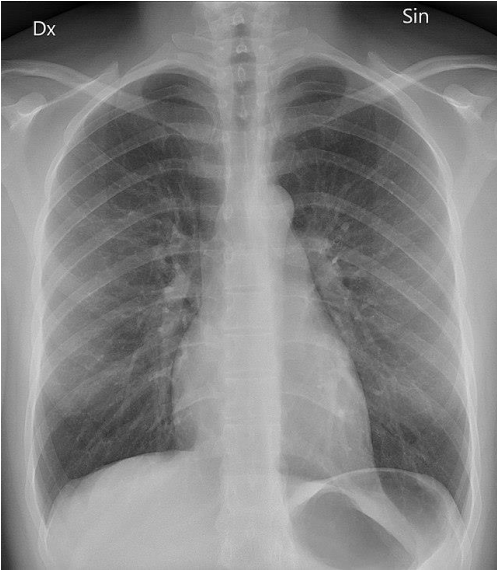
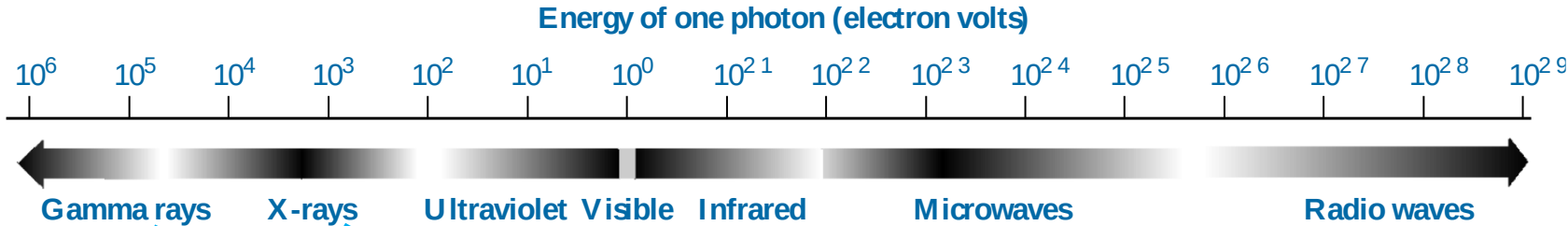


FIGURE 1.18 Images of the Crab Pulsar (in the center of images) covering the electromagnetic spectrum. (Courtesy of NASA.)

Multispectral means Multiview

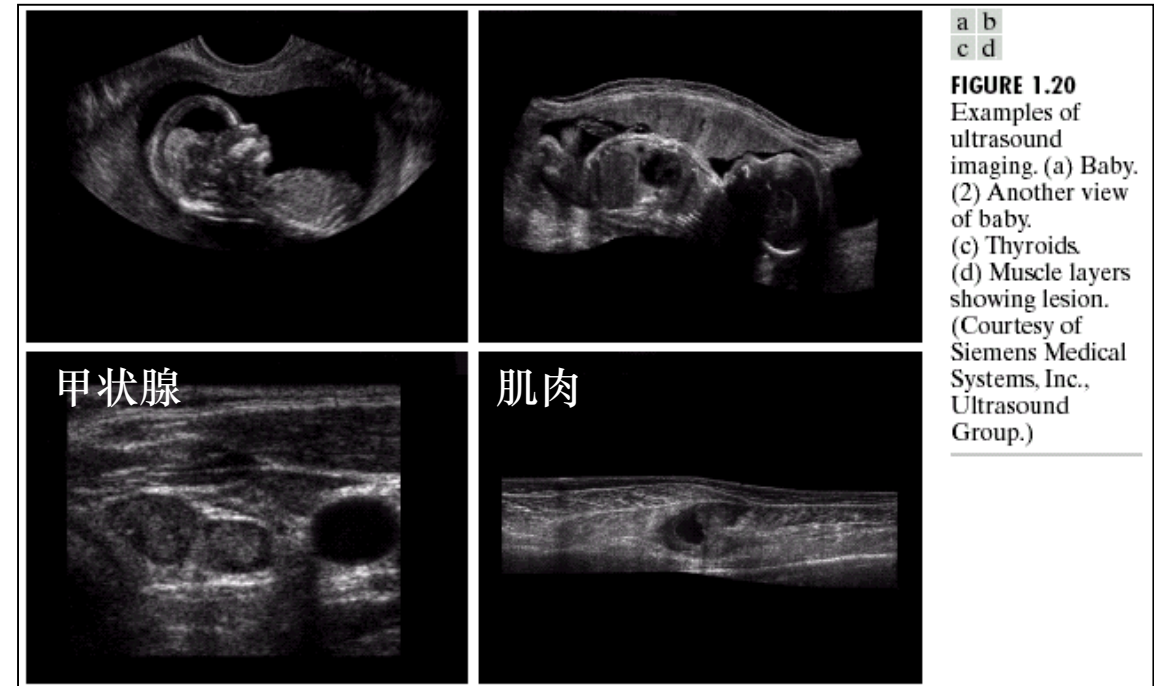
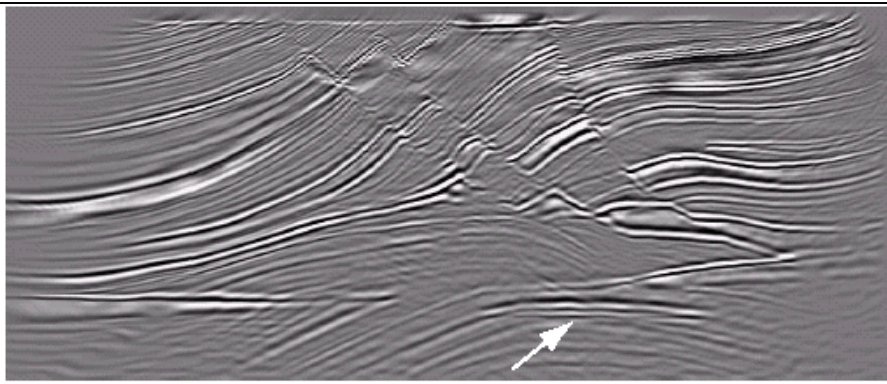
From Gamma to Radio: Medical Images



Images from Other Modalities

- Imaging using “sound” finds application in geological exploration, industry and medicine

FIGURE 1.19
Cross-sectional image of a seismic model. The arrow points to a hydrocarbon (oil and/or gas) trap. (Courtesy of Dr. Curtis Ober, Sandia National Laboratories.)



Images from Other Modalities

- Computer generated Images
- Parameter maps or reconstructed images

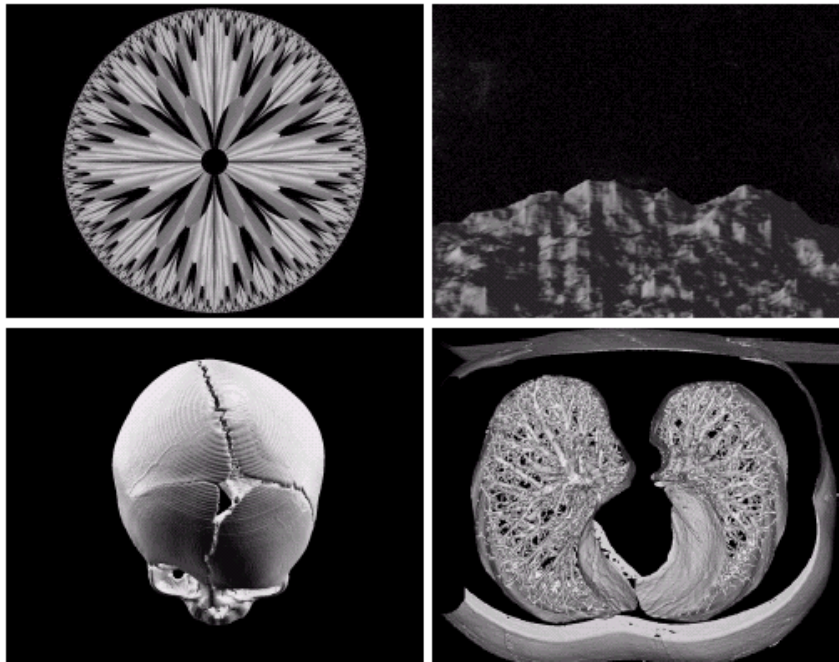
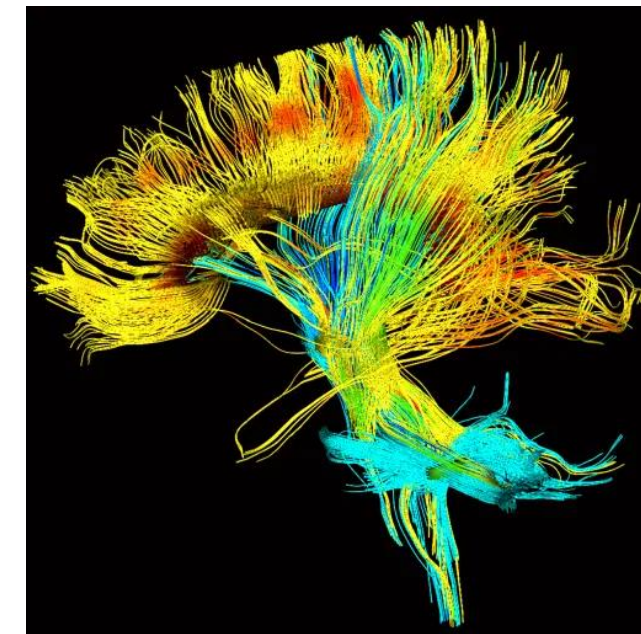


FIGURE 1.22
(a) and (b) Fractal images. (c) and (d) Images generated from 3-D computer models of the objects shown. (Figures (a) and (b) courtesy of Ms. Melissa D. Binde, Swarthmore College, (c) and (d) courtesy of NASA.)



Contents

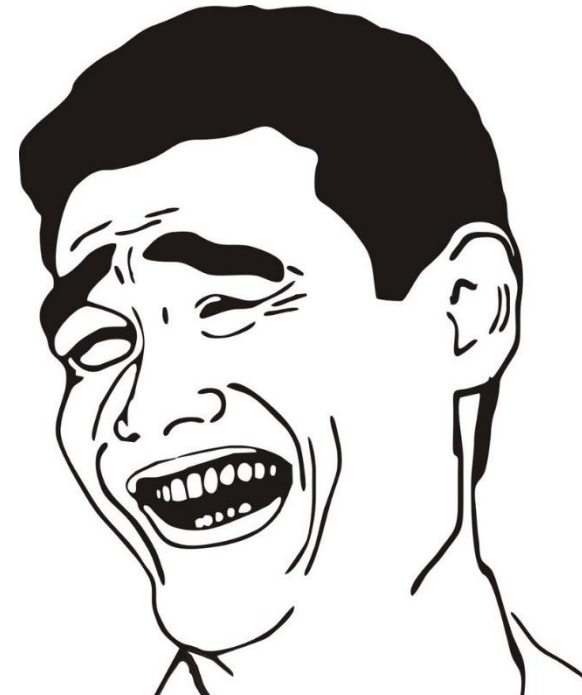
- **Digital Images and Biomedical Images**
- **Format of Digital Images**
 - Types of digital images
 - Storage format of digital images
 - Storage format of medical images
- **What is Image Processing**
- **Human Vision System**

Types of Digital Images

Bitmap

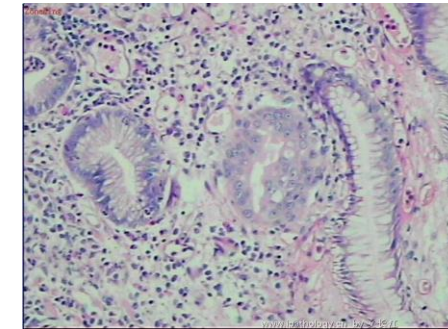
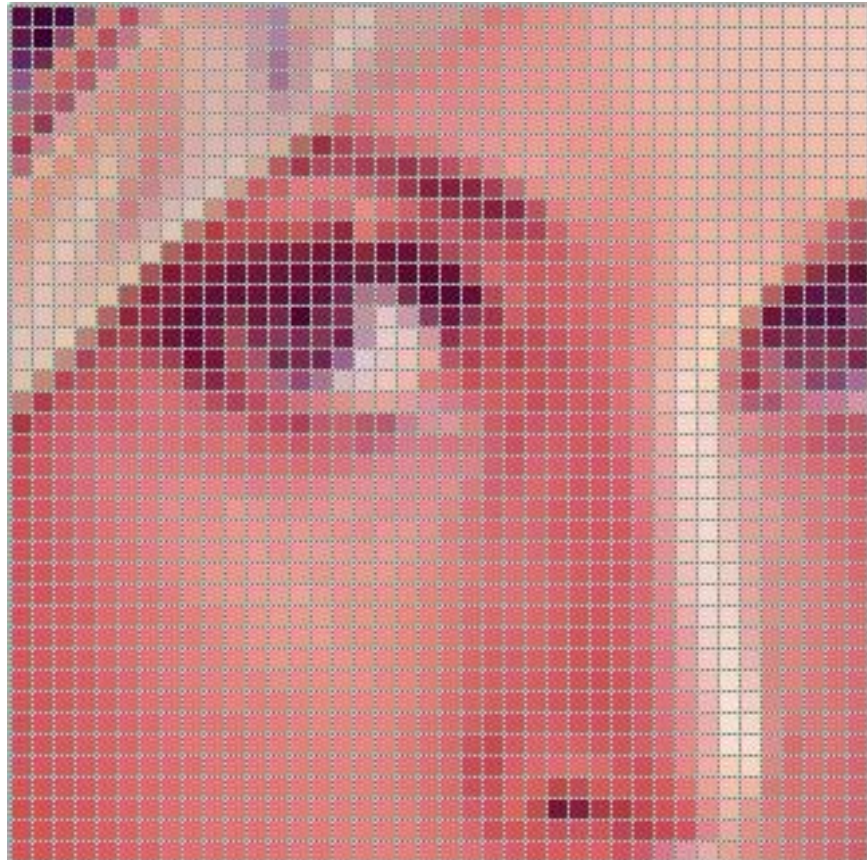


Vector Image



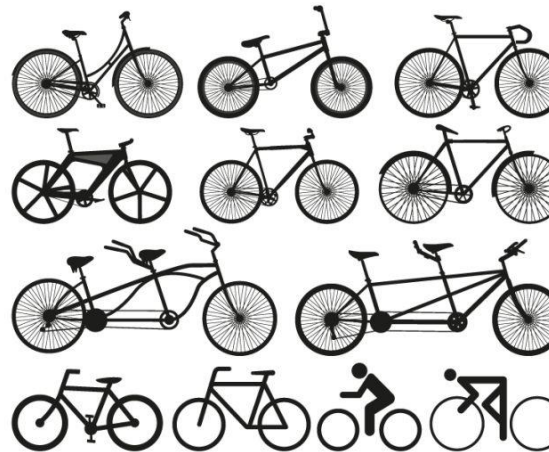
Types of Digital Images

Bitmap: An image represented by many pixels, and each pixel has a color attribute and a position attribute. Bitmaps can be divided into **intensity images, indexed color images and true color images.**



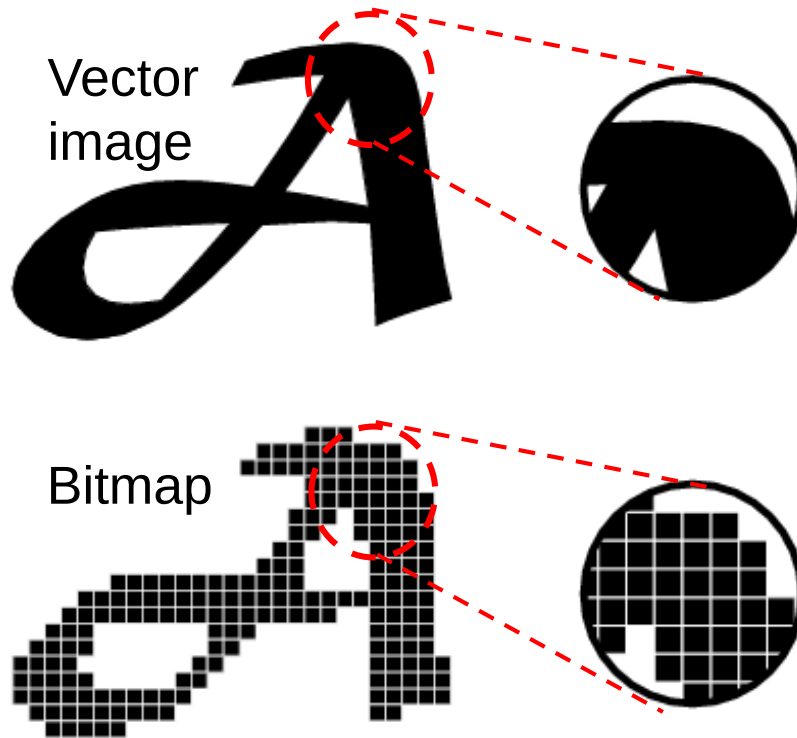
Types of Digital Images

Vector Images: Use geometric shapes such as **points, line segments, and curves** to describe the outline of the image, and **fill the image with color** according to the outline to describe the image. Vector graphics use **vectors** (vectors) to describe the direction of edges when describing the outline of an image



Types of Digital Images

Differences:

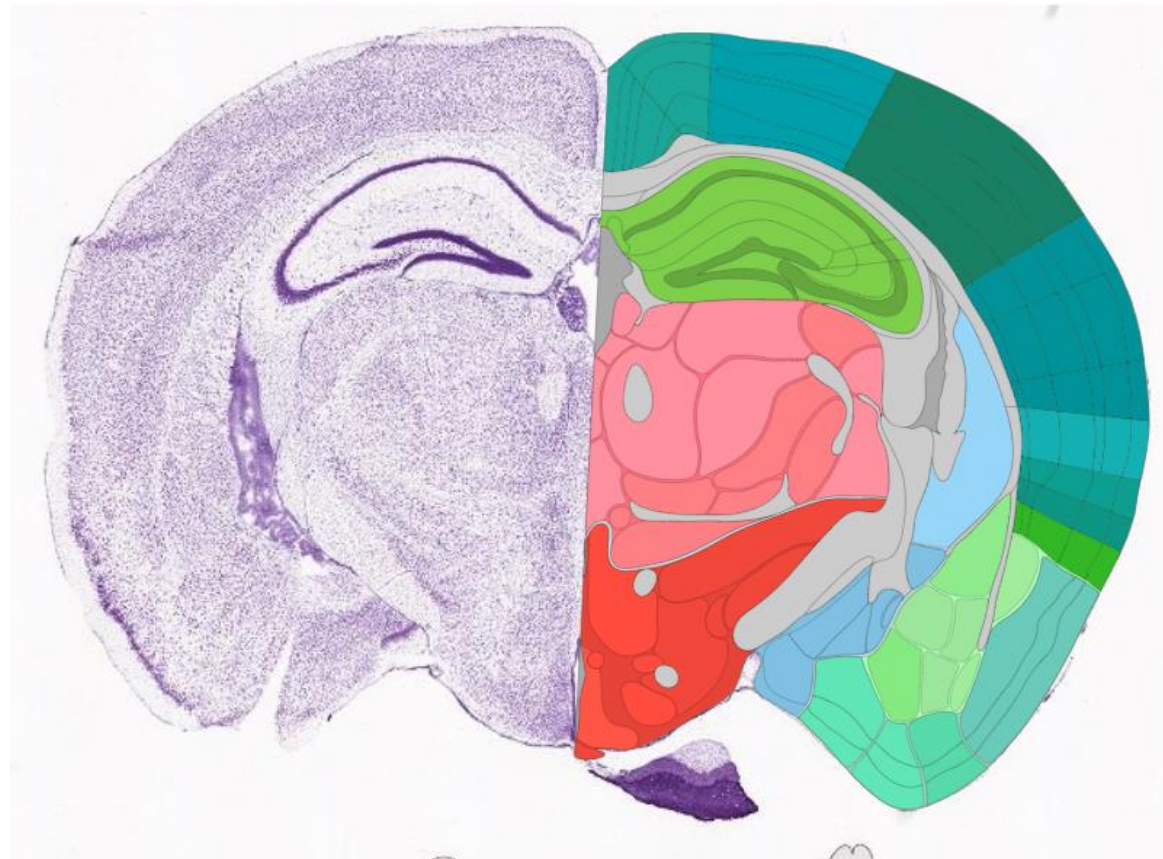


- ❖ Because vector graphics use geometric figures to describe the outline, the image can **be infinitely enlarged** and still displayed clearly.
- ❖ Vector graphics are not easy to produce images with rich tones or too many color changes, such as natural scene images.
- ❖ Bitmaps use pixels to form images, and are suitable for presenting images with rich colors. **Medical images are basically bitmaps.**
- ❖ Bitmap files are generally larger than vector graphics.

Types of Digital Images

Bitmap

Vector Image



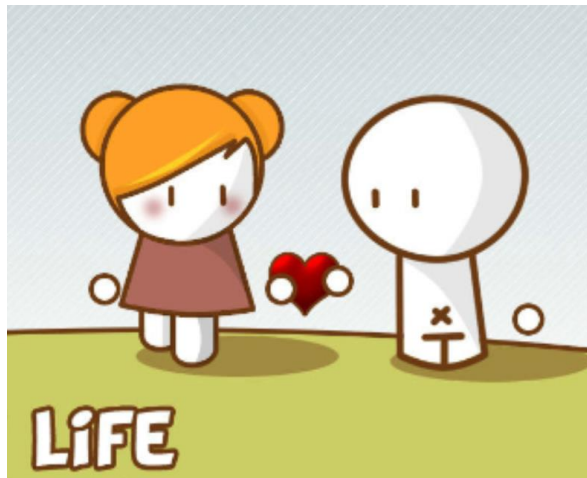
Storage format of Digital Images

- BMP (Uncompressed)
 - Each pixel position corresponds to the spatial position of the pixel in the image, and the bitmap data value corresponds to the brightness value of the corresponding pixel.
- TIFF (TagImage FileFormat)
 - Support a variety of encoding methods, including RGB uncompressed, RLE compressed and **JPEG compressed**, etc.
 - A TIFF image consists of three data structures, **the file header**, one or more directories called IFDs containing **marker pointers**, and **the data itself**.
- GIF (Graphics Interchange Format)
 - Continuous tone lossless compression format based on LZW algorithm, its compression rate is generally about 50%, and it is often used in web animation.

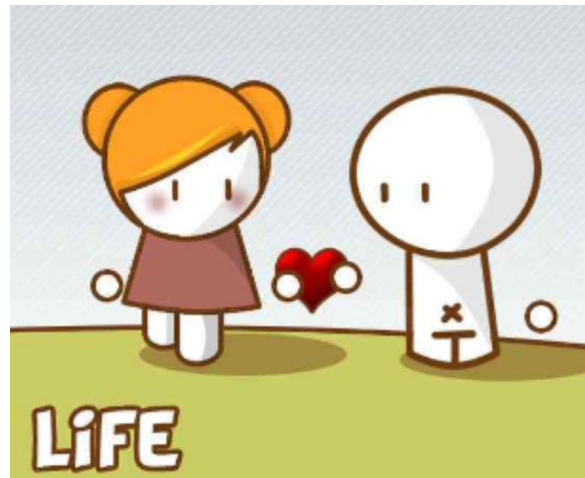
Storage format of Digital Images

❖ JPG (Joint Photographic Experts Group)

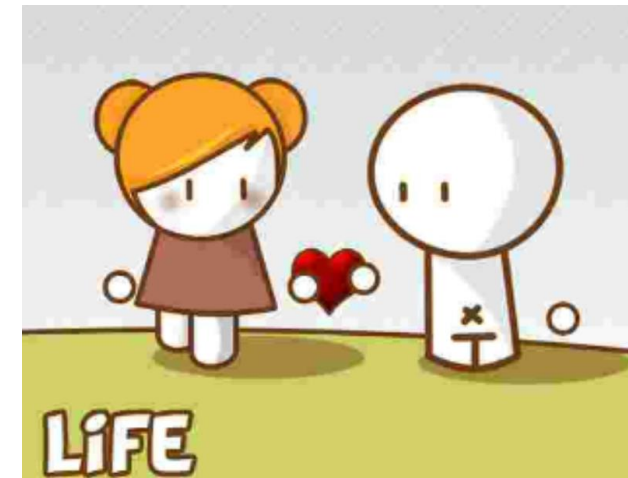
One of the most popular standard formats for high-efficiency **still image** compression. JPEG reduces the file size through compression coding when saving image files, and **deletes image information that is not important to visual cognition**, so as to save storage space and transmission traffic.



Compressing rate
100%



Compressing rate
30%



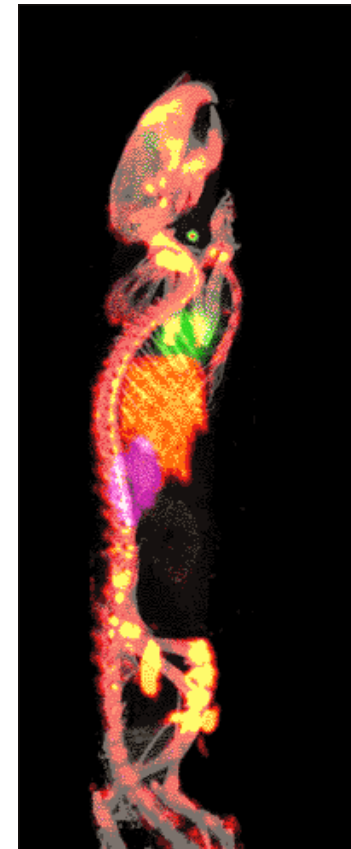
Compressing rate
10%

Storage format of Digital Images

❖ GIF (Graphics Interchange Format)

Continuous tone lossless compression format based on LZW algorithm, its compression rate is generally about 50%, and it is often used in web animation.

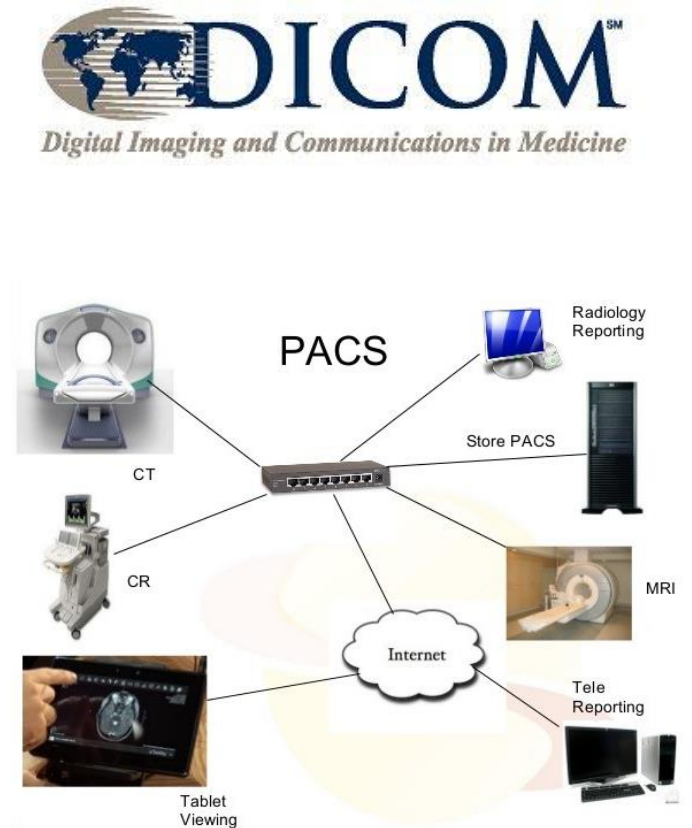
GIF Animation:



Storage format of Medical Images

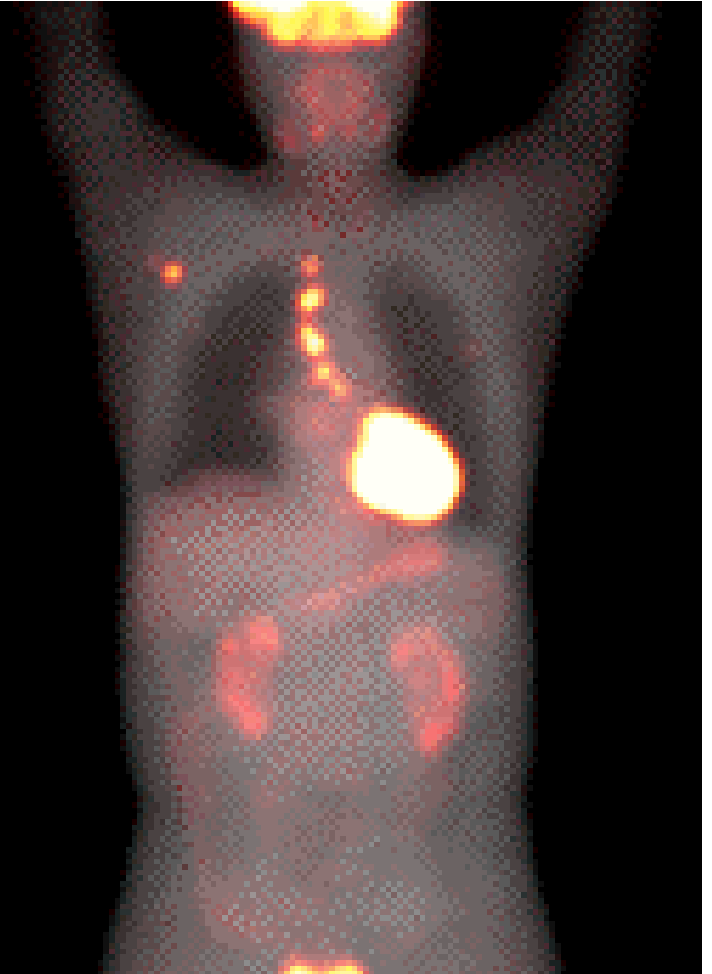
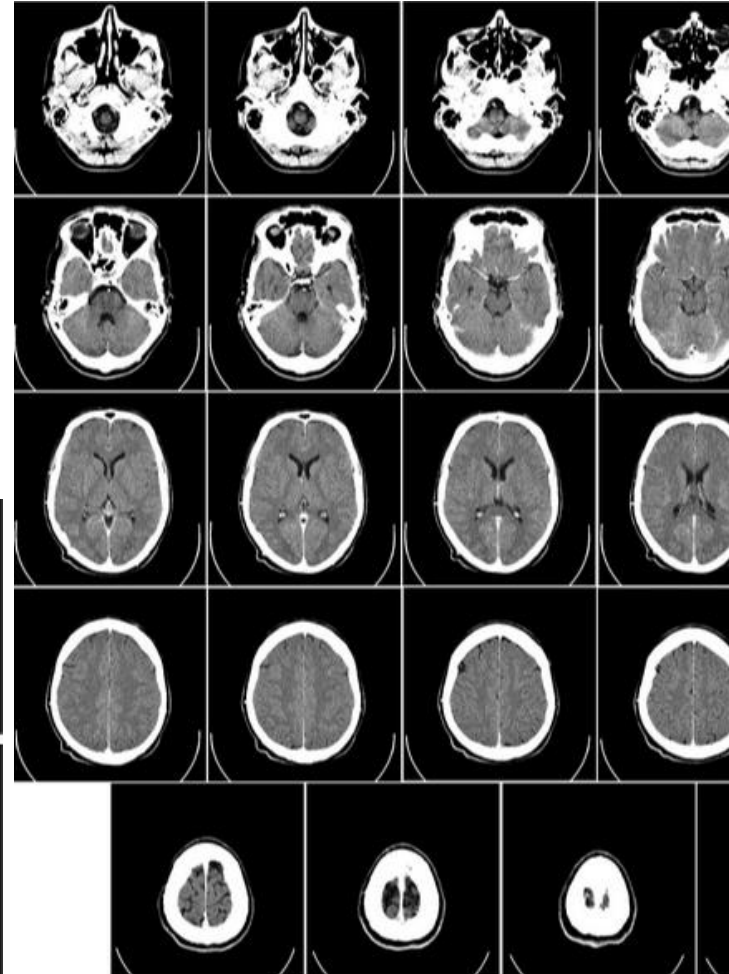
❖ DICOM

- Digital Imaging and Communications in Medicine
- An international standard for medical images and related information (ISO 12052)
- Defines a medical image format that can be used for data exchange with a quality that meets clinical needs.
- Widely used in medical imaging (X-ray, CT, magnetic resonance imaging, ultrasound, etc.), and also well accepted in other medical fields such as ophthalmology and dentistry.



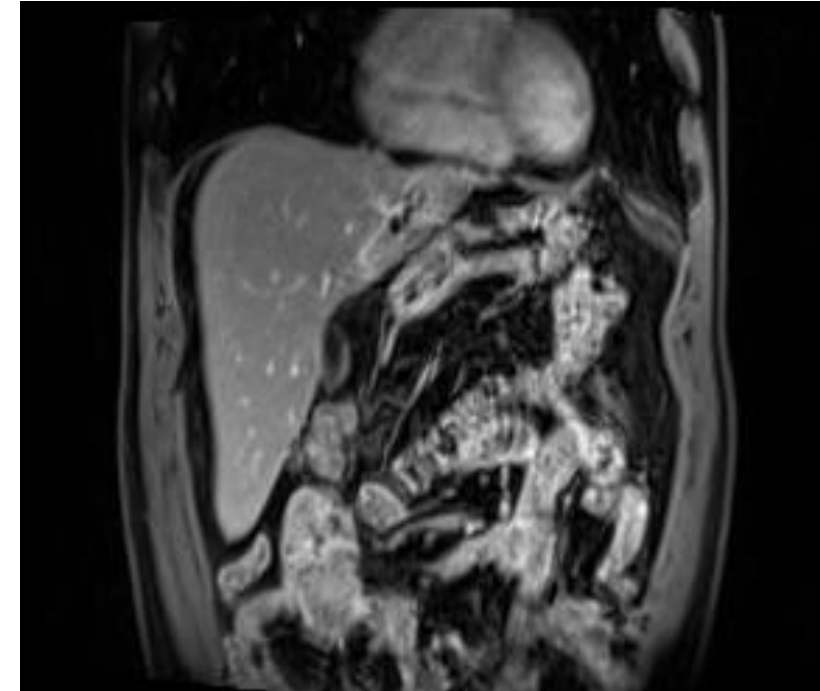
Storage format of Medical Images

DICOM



DICOM Header

- dicomInf =
- Filename: [1x104 char]
- FileModDate: '27-11月-2021 09:53:10'
- FileSize: 125456
- Format: 'DICOM'
- FormatVersion: 3
- Width: 288
- Height: 200
- BitDepth: 12
- ColorType: 'grayscale'
- FileMetaInformationGroupLength: 200
- FileMetaInformationVersion: [2x1 uint8]
- MediaStorageSOPClassUID: '1.2.840.10008.5.1.4.1.1.4'
- MediaStorageSOPInstanceUID: [1x52 char]
- TransferSyntaxUID: '1.2.840.10008.1.2.4.70'
- ImplementationClassUID: '1.2.826.0.1.3680043.1.1.4.3.82.2'
- ImplementationVersionName: 'DCMOBJ4.3.82.2'
-



Storage format of Medical Images

❖ NIfTI (Neuroimaging Informatics Technology Initiative)

- The file is generally **made up of a header followed by volumetric data**.
- The format supports up to **5 dimensions**, each value in the volume can range from unsigned characters up to long double precision numbers, and includes support for r,g,b volumes.
- The NIfTI format is adapted from **Analyze™ 7.5 (.img/.hdr)**, developed by Biomedical Imaging Resource (BIR) at Mayo Clinic. NIfTI and Analyze 7.5 are still compatible – NIfTI simply adds additional fields.
- Compressed format of NIfTI: .nii.gz

Contents

- Digital Images and Biomedical Images
- Format of Digital Images
- What is Image Processing
- Human Vision System

What is Image Processing?

- Takes one image as input and produces a modified version of that image. Thus, the input/output are all images
- No clear-cut boundaries between image *processing* and *analysis*

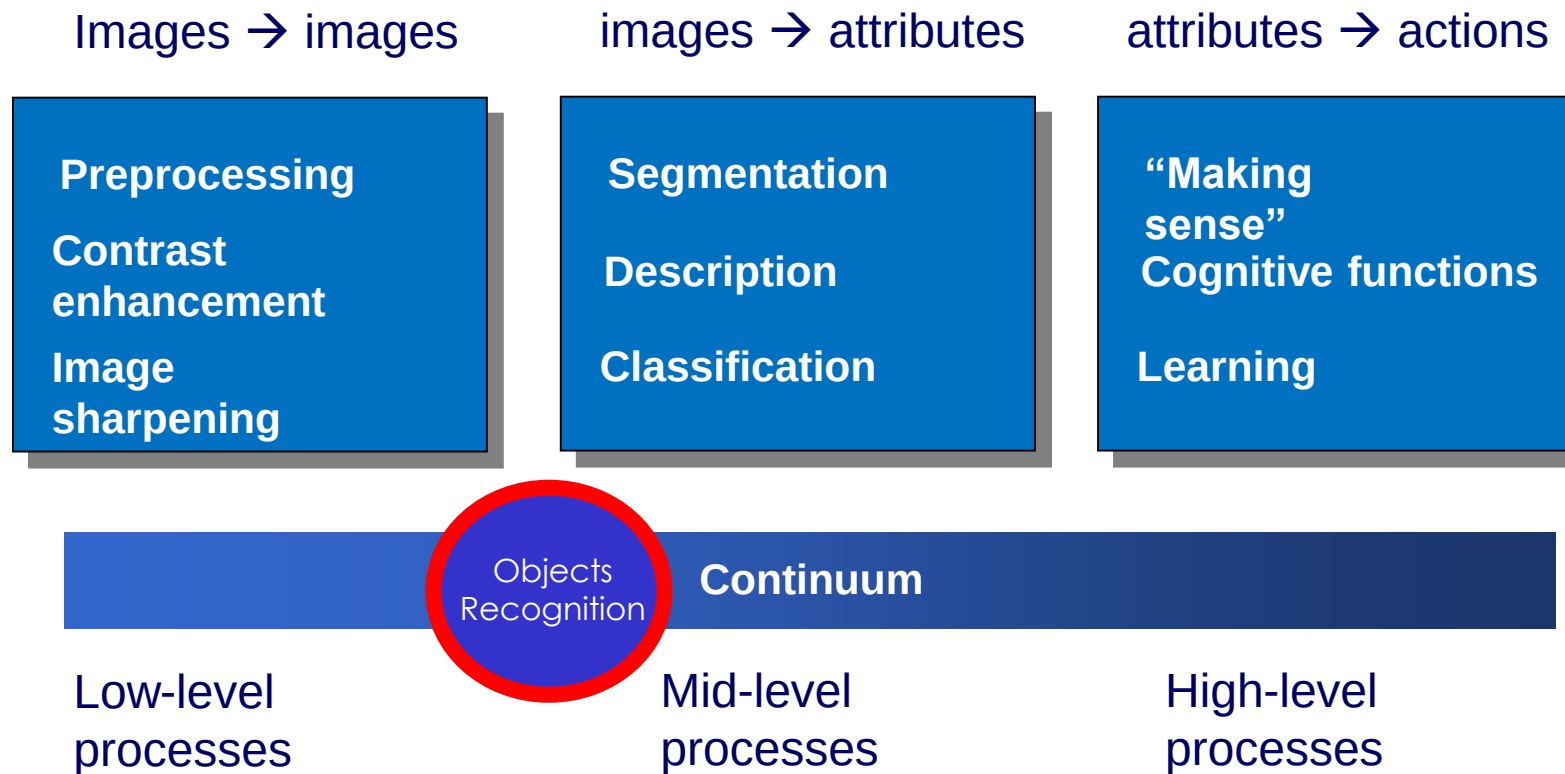
Image Processing

→ Image Analysis (understanding)

→ Computer Vision (a branch of AI)

What is Image Processing?

A better view

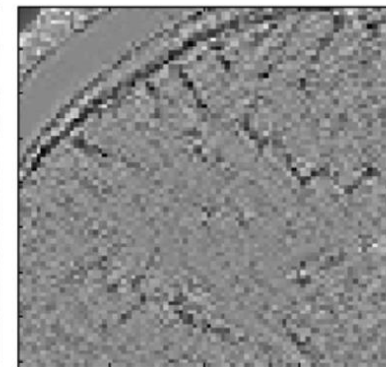
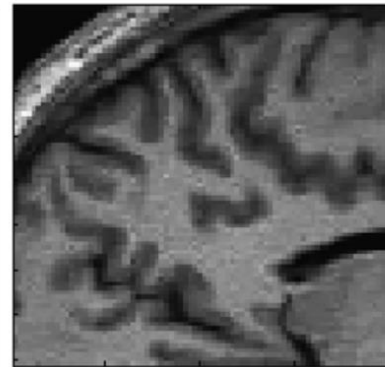
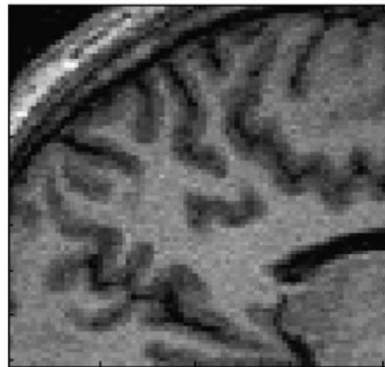
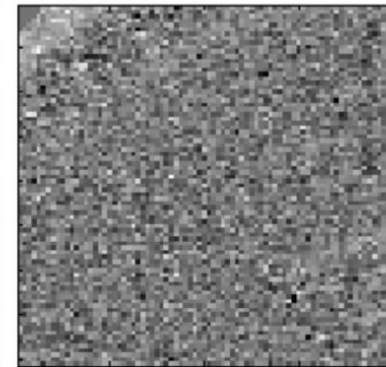
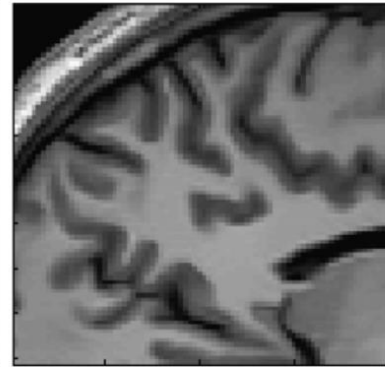
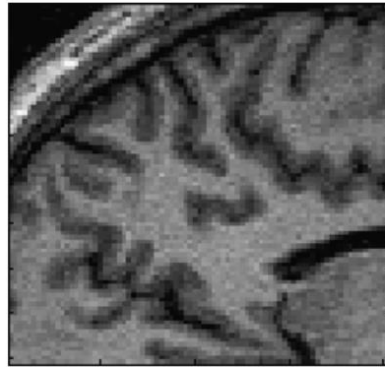


Example of BIP: De-noising

Noise image

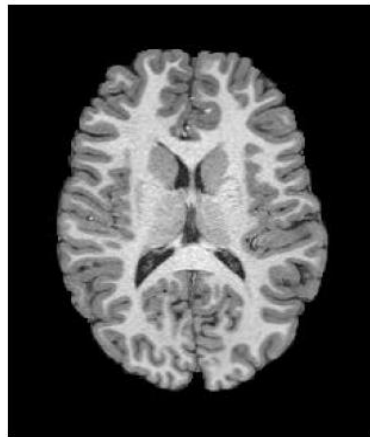
Denoised image

Residuals

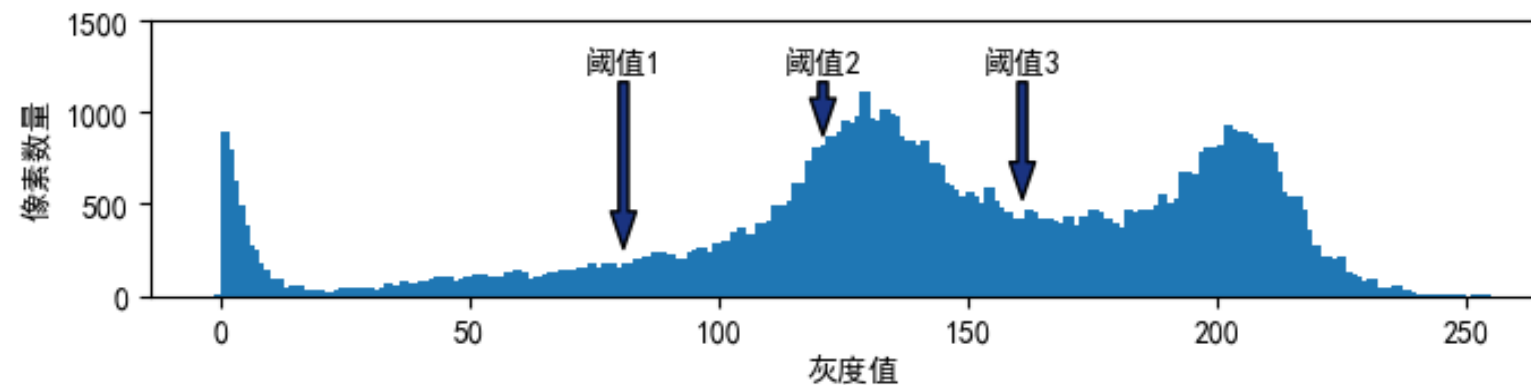
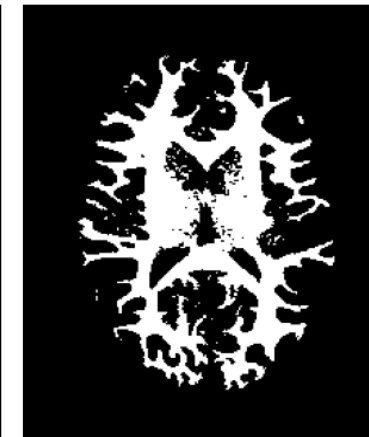
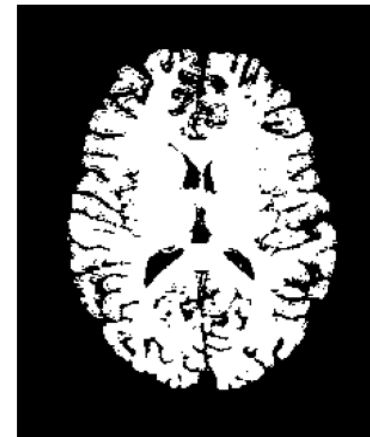


Example of BIP: Image Thresholding

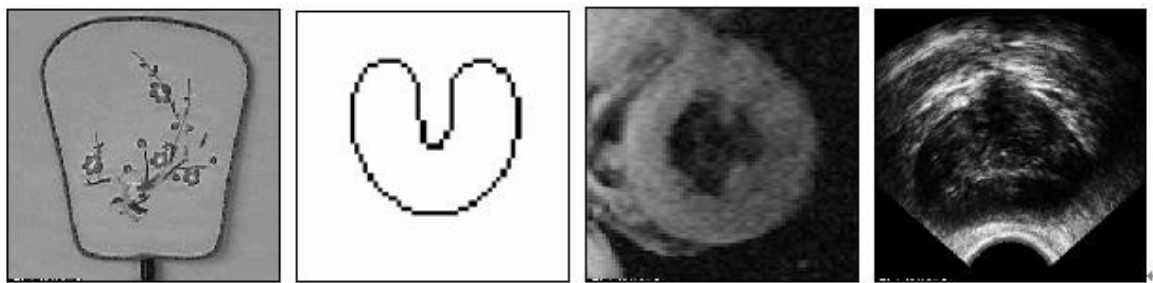
Original image



Segmented image



Example of BIP: Boundary detection

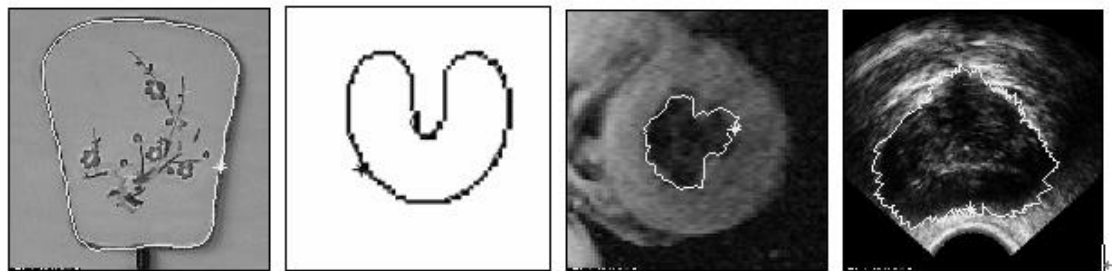


(a)

(b)

(c)

(d)

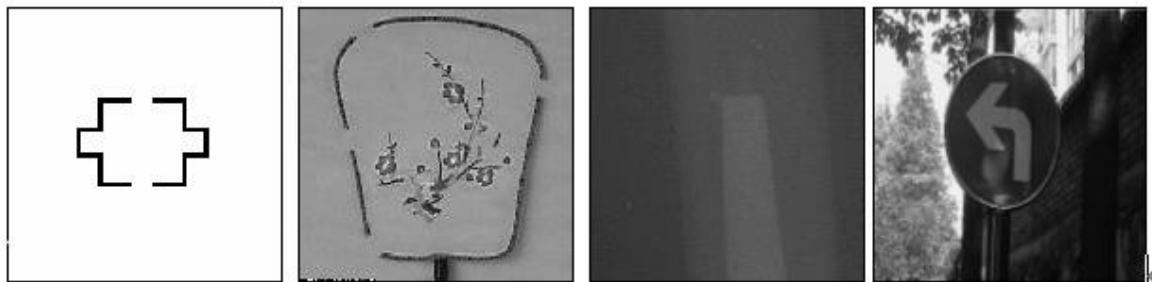


(i)

(j)

(k)

(l)

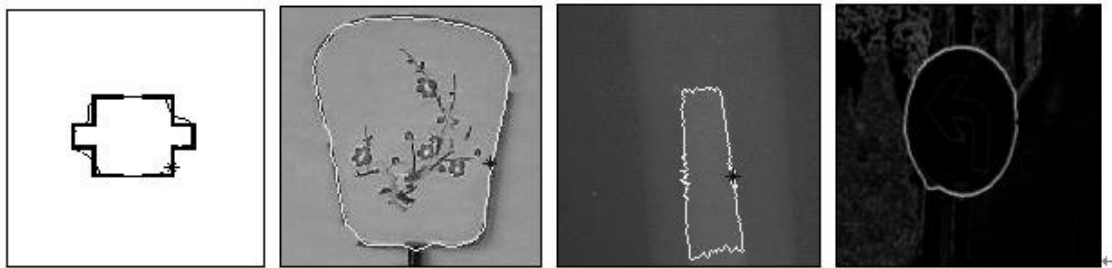


(e)

(f)

(g)

(h)



(m)

(n)

(o)

(p)

Example of BIP: Image Segmentation

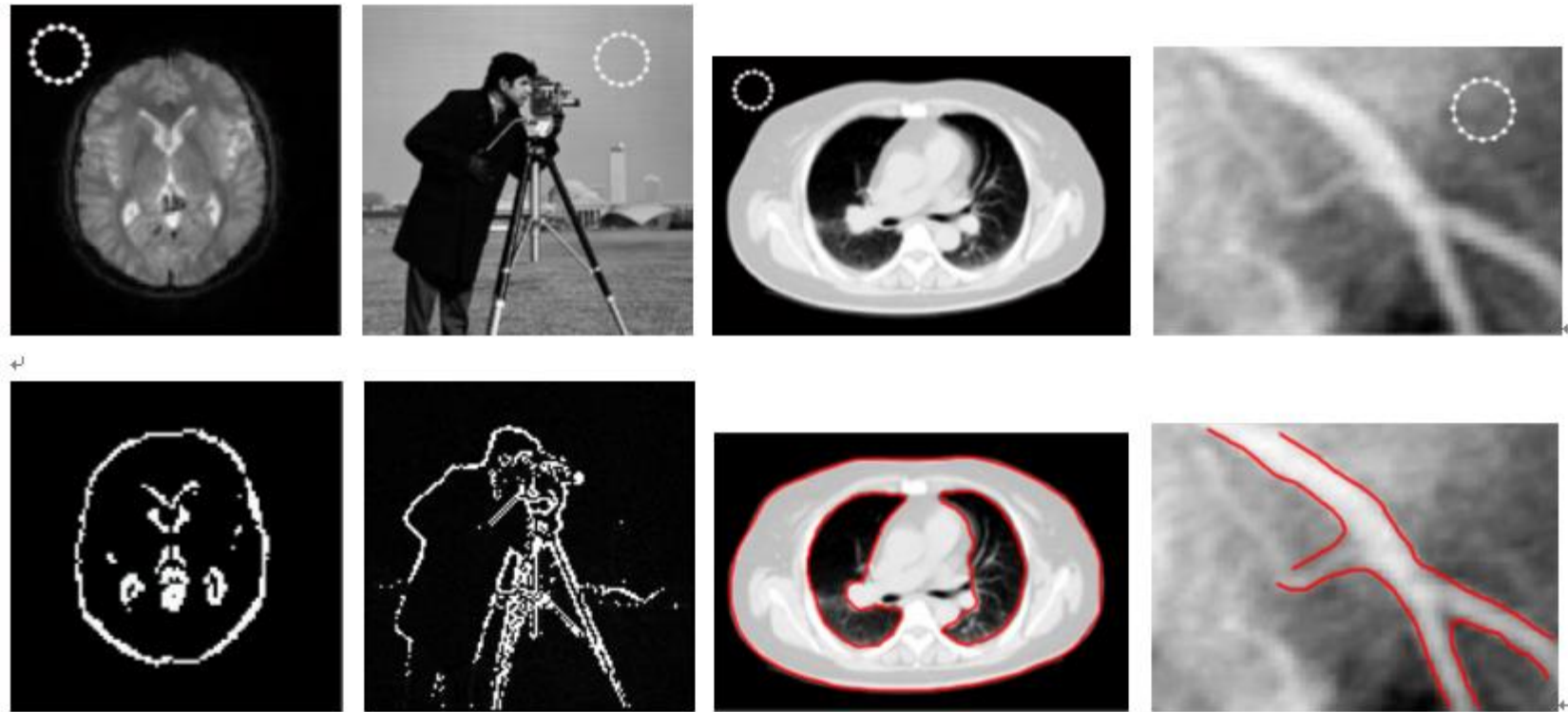
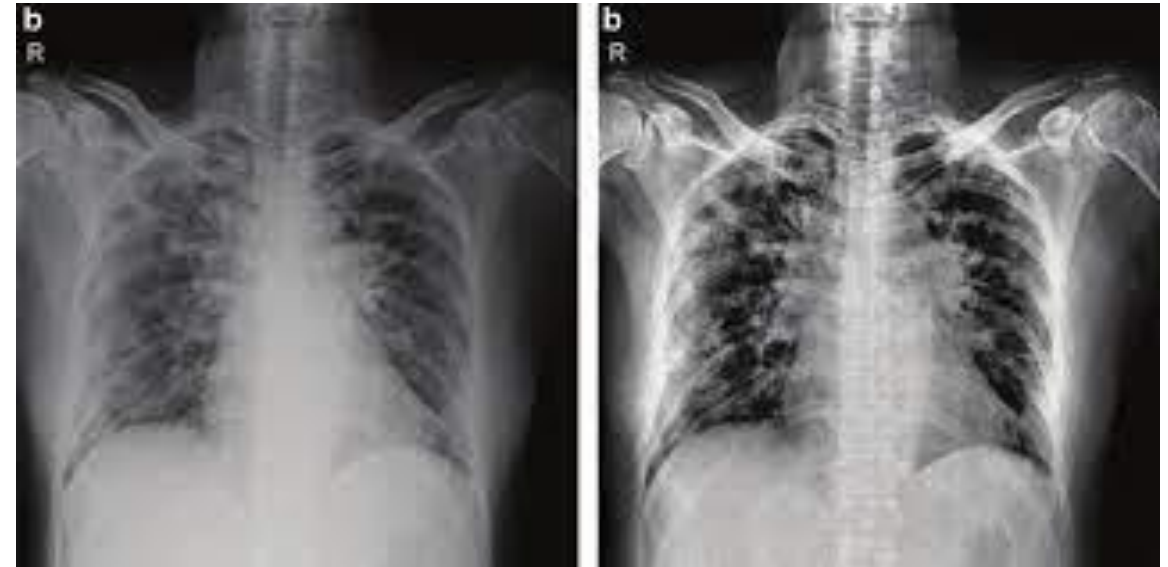
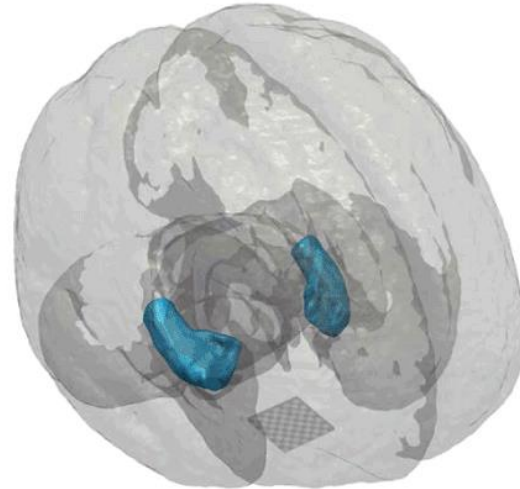


Fig. 10. Edge detection and boundary extraction with additional real images. First row: original images; Second row: the detected edges and the extracted boundaries (in the red curves) using the QSDM.⁴

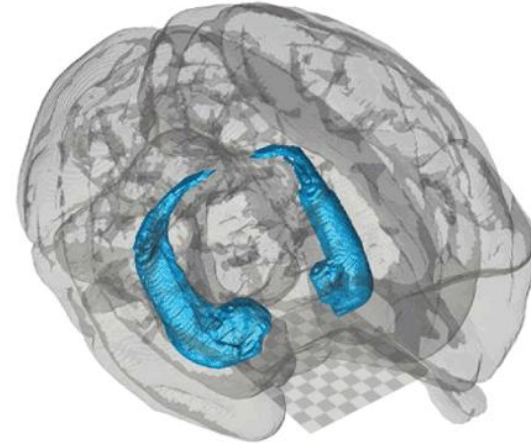
Example of BIP: Image enhancement



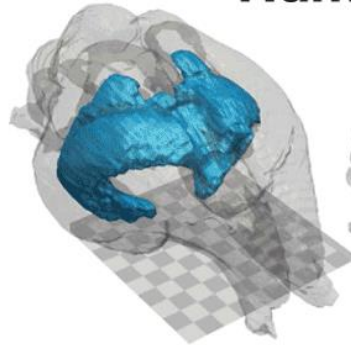
Example of BIP: Volume rendering



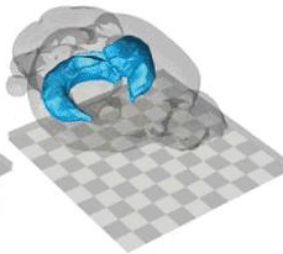
Human



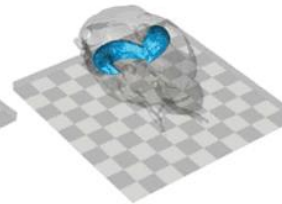
Macaque



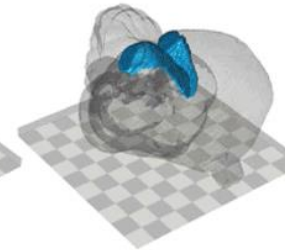
Rabbit



Rat

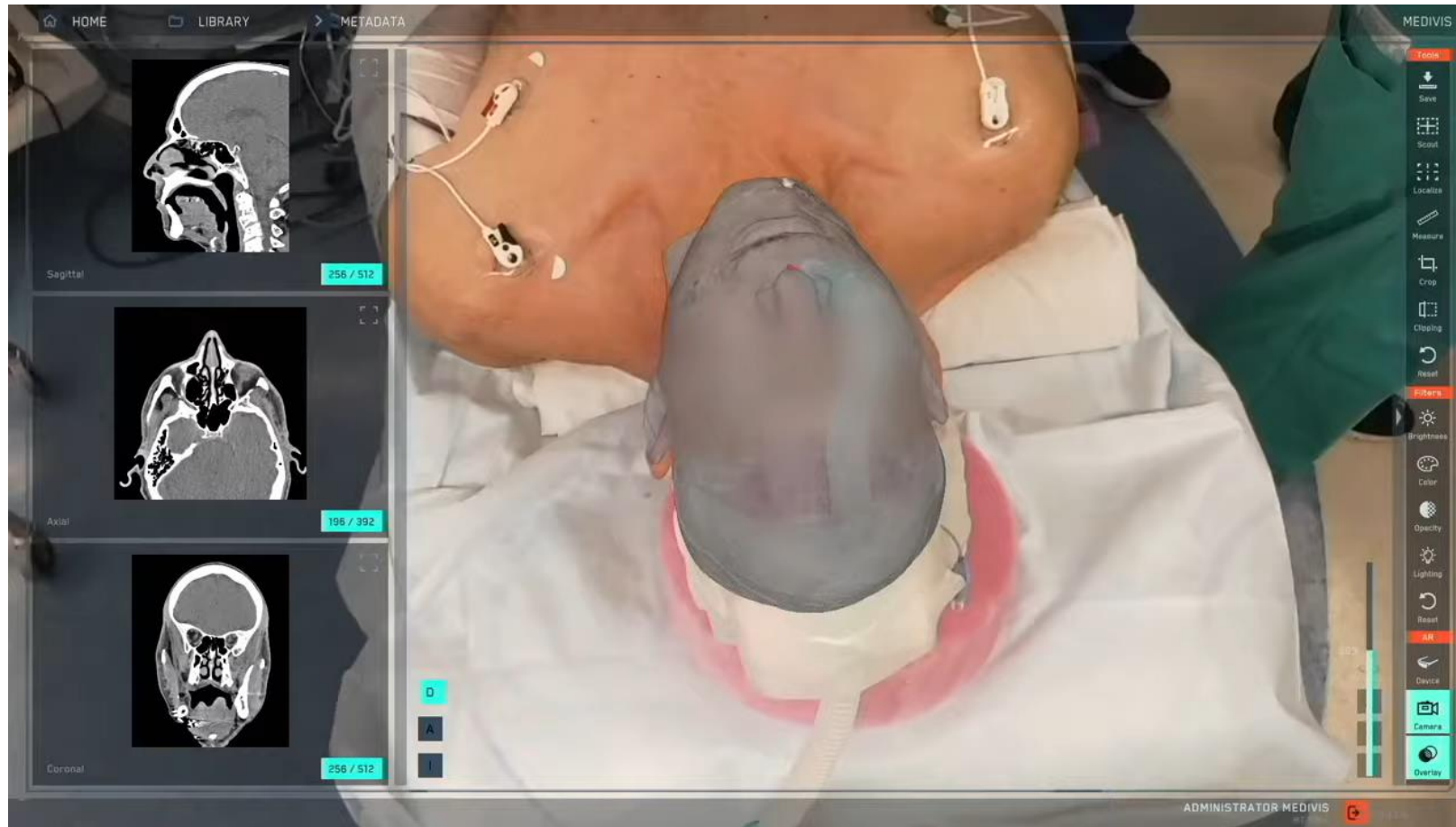


Mouse



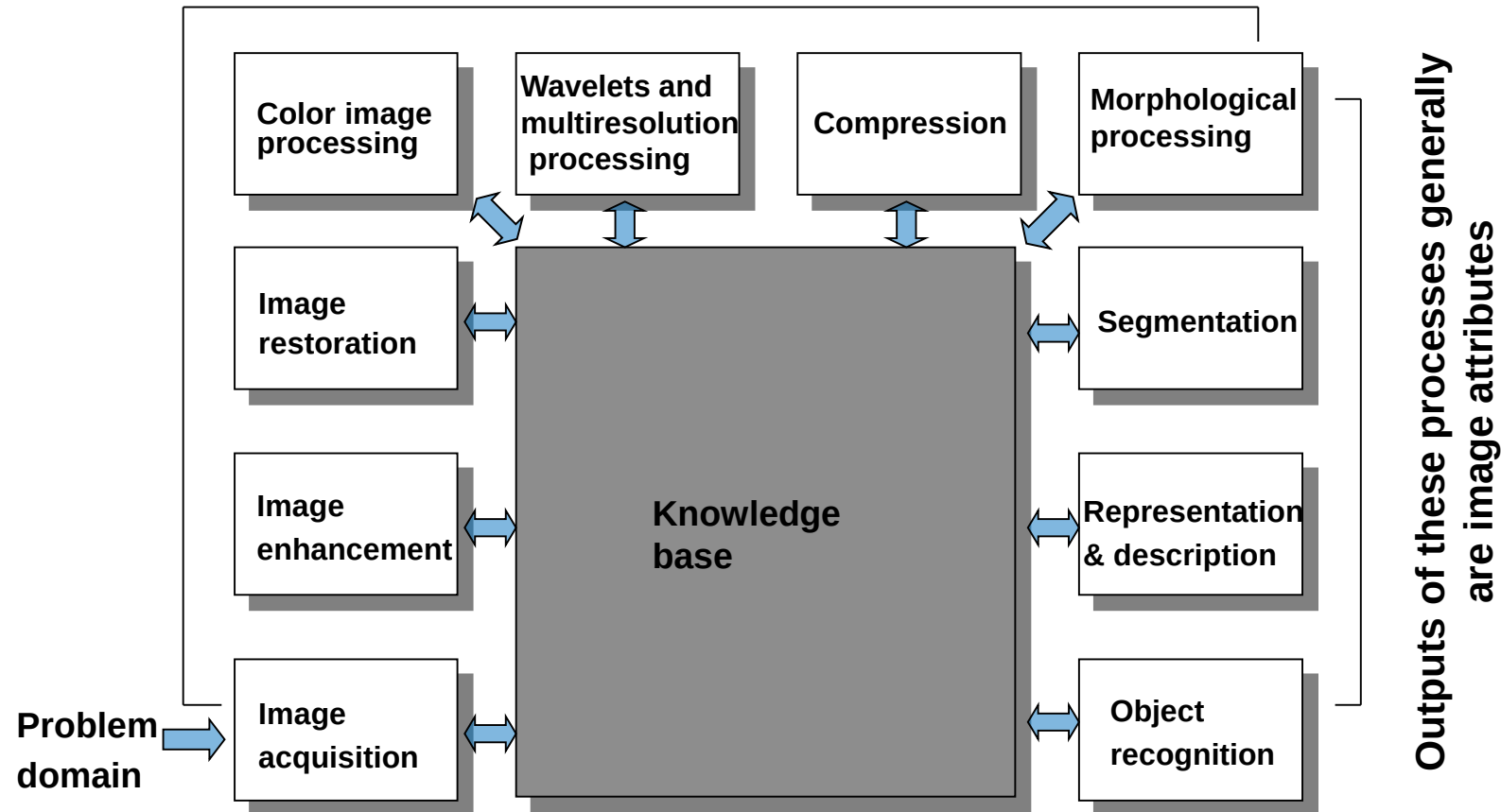
Pigeon

Example of BIP: Virtual reality



Fundamental Steps in BIP

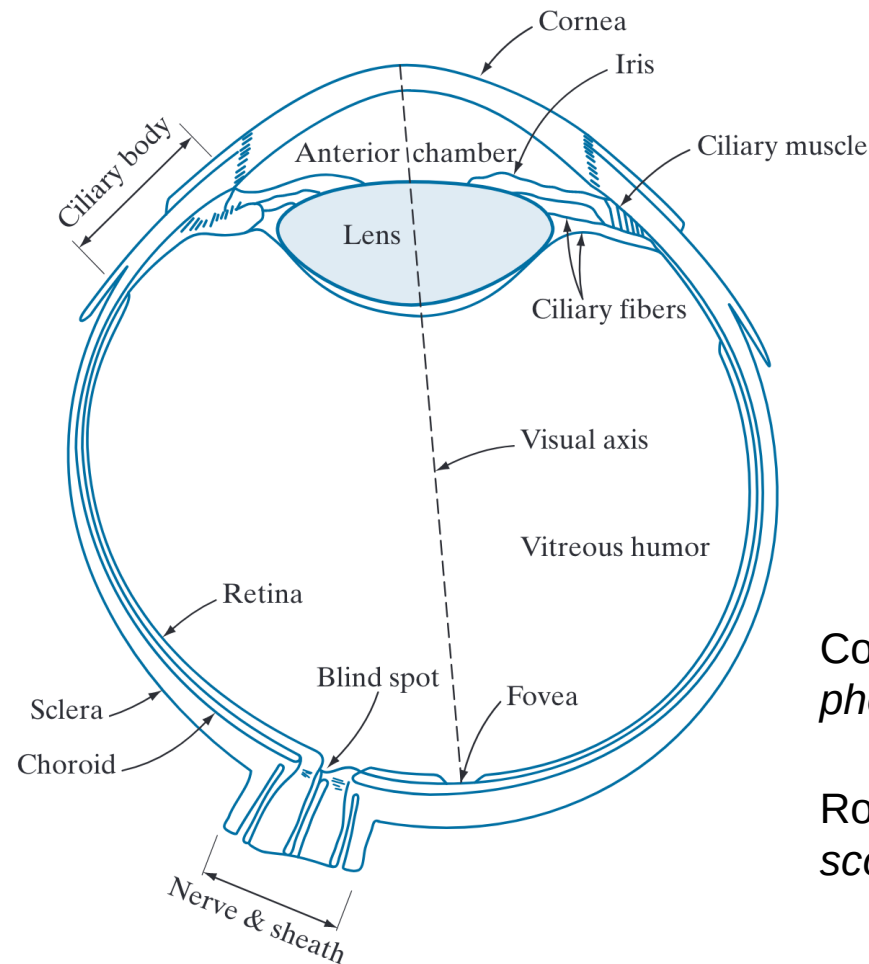
Outputs of these processes generally are images



Contents

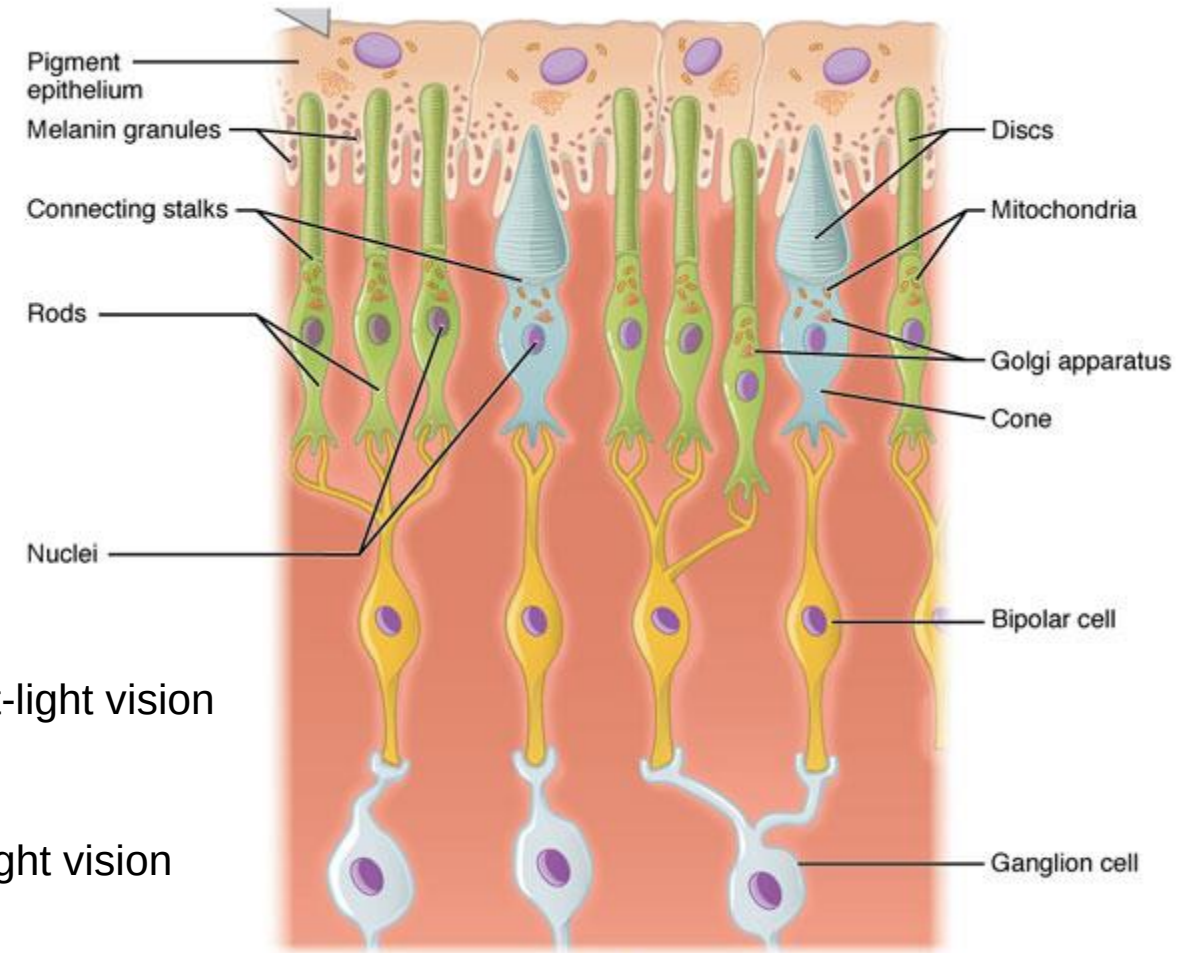
- **Digital Images and Biomedical Images**
- **Format of Digital Images**
- **What is Image Processing**
- **Human Vision System**

Elements of Visual Perception



Cones: 6-7M
photopic or bright-light vision

Rods: 75-150M
scotopic or dim-light vision

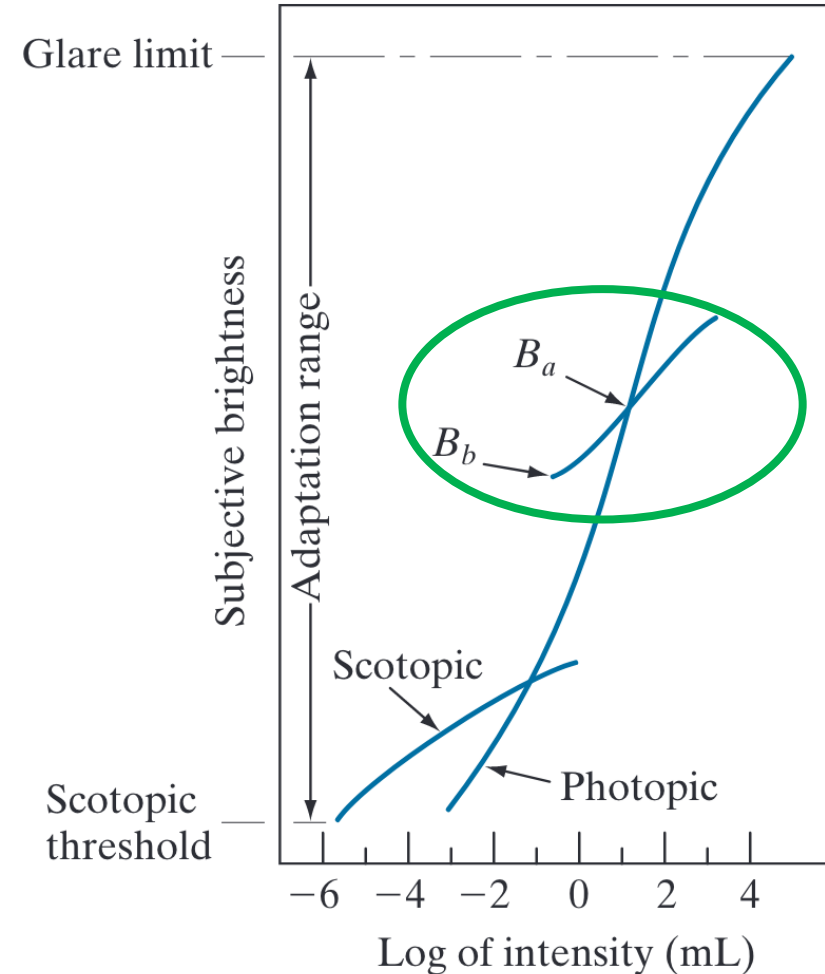


Brightness Adaptation and Discrimination

The range of light intensity levels to which the human visual system can adapt is on the order of 10^{10}

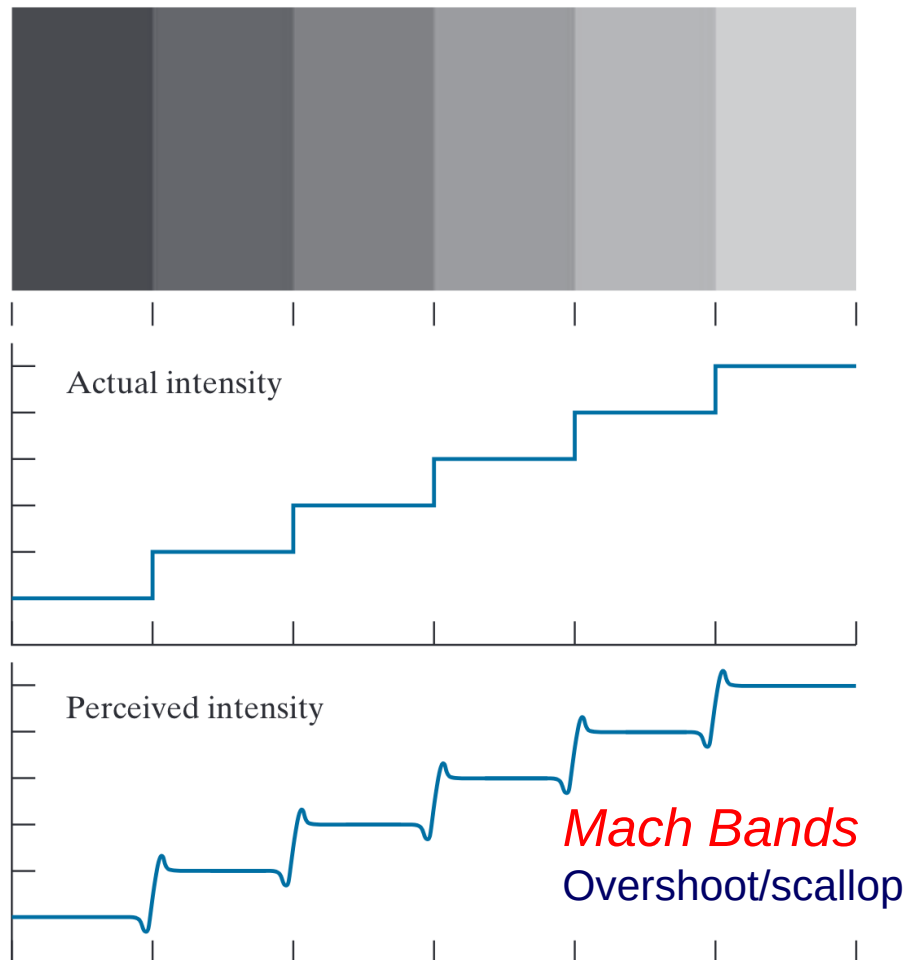
Experimental evidence indicates that *subjective brightness* (intensity as *perceived* by the human visual system) is a **logarithmic function** of the light intensity incident on the eye.

It accomplishes this large variation by changes in its overall sensitivity, a phenomenon known as **brightness adaptation**.



Objective vs Subjective

- The perceived brightness is not a simple function of intensity.

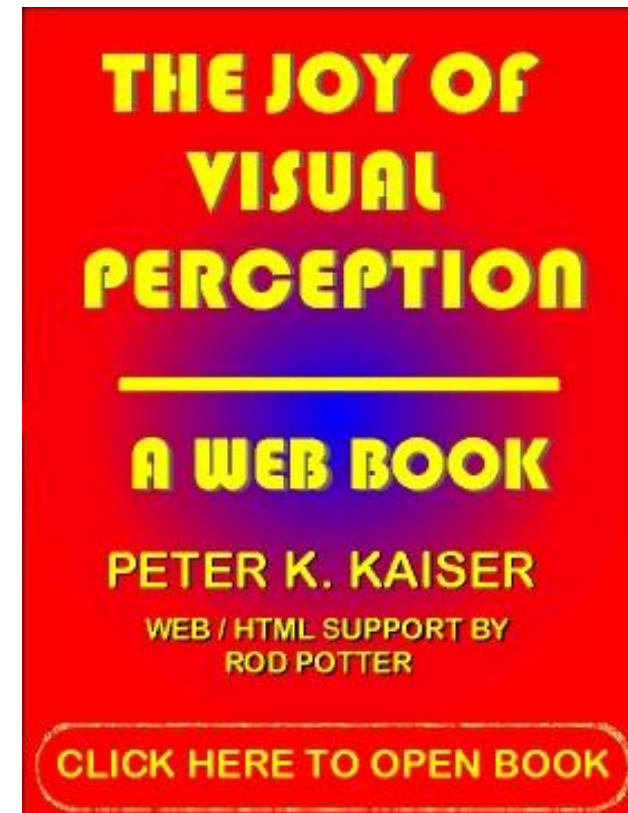
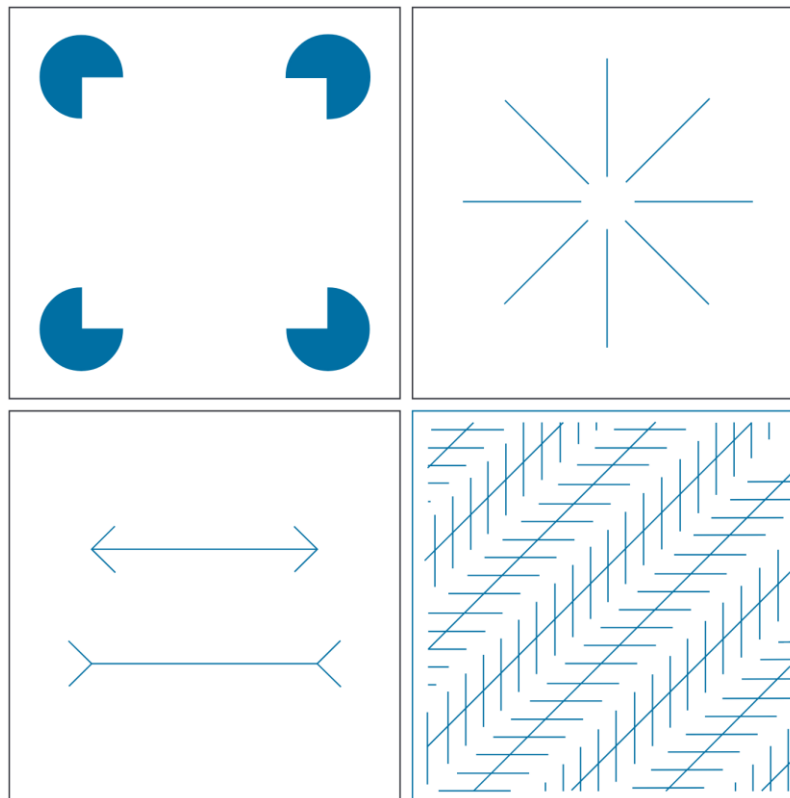


a b c

FIGURE 2.8 Examples of simultaneous contrast. All the inner squares have the same intensity, but they appear progressively darker as the background becomes lighter.

What's Wrong with our Eyes?

- **Optical illusions**, in which the eye fills in nonexisting information or wrongly perceives geometrical properties of objects.



Summary

- Digital image
- Digital image format
- Digital image processing
- Human eye: Physiology and function

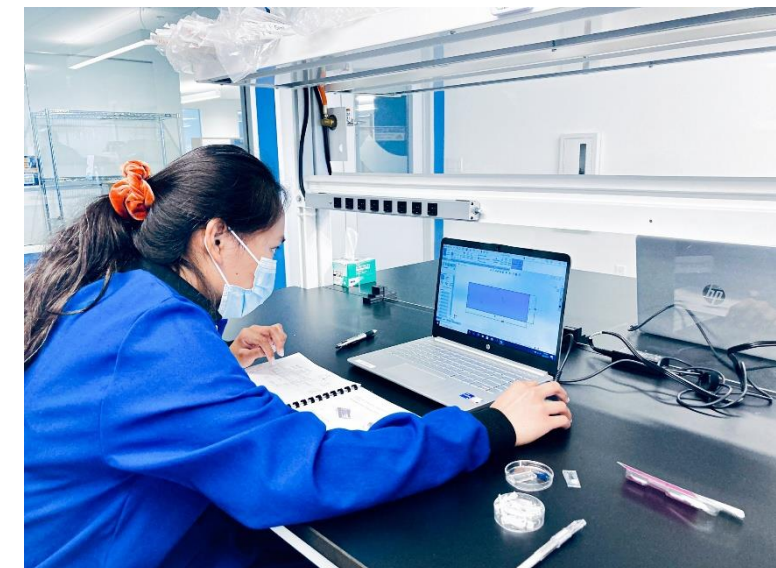
Any Questions?

Coming Lab

- Lab is the most important part!
- Your Computer
- Coding enviroment
- Program



wet



dry